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SURVEY

Application of Artificial Intelligence for Classification, Segmentation, Early Detection, Early Diagnosis, and Grading of Diabetic Retinopathy From Fundus Retinal Images: A Comprehensive Review

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ABSTRACT Diabetic Retinopathy (DR) remains a major factor contributing to vision loss worldwide, particularly among individuals with diabetes. Timely and accurate diagnosis of DR is essential to prevent vision loss and guide clinical intervention. In recent decades, Artificial Intelligence (AI) has demonstrated remarkable potential in revolutionizing DR disease identification and risk assessment. This review paper aims to perform the comprehensive analysis regarding the present status of research and developments in the field of DR using AI technologies. This comprehensive review emphasize the five categories of DR applications such as classification, detection, early diagnosis, segmentation and grading the severity of DR's. This review article presents how AI technologies are leveraged in the current research works considering all these five DR applications. This is illustrated by using the AI-based DR workflow. It highlights the few significant case studies utilizing the AI technologies in real life problem-solving. Further, the adoption of Explainable Artificial Intelligence (XAI) techniques in the field of DR applications are discussed for model generalization and clinical validation. Finally, it presents the available benchmarking datasets, performance metrics, key challenges and future scope for DR research utilizing the AI technologies. This review work unveils several challenges and limitations with respect to input data modality, data preprocessing and data-fusion drawing the scope for further enhancements that can be made for a better DR prediction and detection.

INDEX TERMS Deep learning, machine learning, explainable AI, image preprocessing, diabetic retinopathy, fundus imaging, optical coherence tomography, optical coherence tomography angiography, transfer learning.

I. INTRODUCTION

Diabetic Retinopathy (DR) is a significant neurovascular complication of diabetes and a major factor leading to vision impairment or blindness, particularly in working-age and elderly people across the world [1]. Recent epidemiological data from the American Academy of Ophthalmology indicates global diabetes rise to over 387 million people, anticipated to increase to over 592 million individuals by 2035, with 93 million individuals affected by DR. This leads to 77.3% of individuals with type 1 diabetes and 25.1% of individuals with type 2 diabetes having the disease, and 25–30% of those people are predicted to develop diabetic macular edema (DME), which can be vision-threatening. Among these DR patients, up to 5% may require vitrectomy surgery, and between 5% and 8% may need laser

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TABLE 1. Glossary table.

Full Form	Abbreviation
Circle Hough Transform	CHT
Dice Coefficient	DICE
Archimedes optimization algorithm	AOA
Area under the Receiver Operating Characteristic Curve	AUROC
Artificial Intelligence	AI
Asia Pacific Teleophthalmology Society	APTOS
Active Deep Learning	ADL
Bahawal Victoria Hospital	BVH
Batch Normalization	BNs
Battle Royale Optimization	BRO
Bidirectional Gated Recurrent Convolutional Unit	BGRCU
Binary Bat Algorithm	BBA
Blood Vessels	BVs
Class Activation Mapping	CAM
Class Attention Learning	CAL
Co-Occurrence Matrix	COM
Color Fundus Photography	CFP
Colour Fundus Photographs	CFPs
Contrast Limited Adaptive Histogram Equalization	CLAHE
Convolutional Neural Network	CNN
Coyote Optimization Algorithm	COA
Deep Capillary Plexus	DCP
Deep Convolutional Neural Network	DCNN
Deep Extreme Learning Machine	DELM
Deep Inductive Transfer Learning	DITL
Deep Learning	DL
Deep Learning System	DLS
Deep Transfer Learning-based Computerized Diagnostic Systems	DTLCDS
Deep Vascular Plexus	DVP
Diabetes Mellitus	DM
Diabetic Macular Edema	DME
Diabetic Macular Oedema	DMO
Diabetic Retionopathy	DR
DR Classification	DRC
Diagnostic Assessment–Based DL for DR	DAD-DR
DL Algorithm -DR	DLA-DR
Elastic Net-Regularized Logistic Regression	LR-EN
Enhanced Super Resolution Generative Adversarial Network	ESRGAN
Equilibrium Optimizer	EO
Explainable Artificial Intelligence	XAI
Explainable Boosting Machine	EBM
eXtreme Gradient Boosting	XGBoost
Extreme Learning Machine	ELM
Exudates	EX
False Positive Rate	FPR
Fractional Calculus	FC
Fractional War Royale Optimization	FrWRO
Fractional War Strategy Optimization	FrWSO
Fully Dense Fusion Neural Network	FD-CNN
Fundus Fluorescein Angiography	FFA
Fundus Images	FI
Freshet Inception Distance	FID
Gabor Filtering	GF
GCA-EfficientNet	GENet
Generative Adversarial Network	GAN
Genetic Algorithm–Based DR	GAB-DR
Global Channel Attention	GCA
Global Information Embedding	GIE
Glowworm Swarm Optimization	GSO
Graphical User Interface	GUI
Graphics Processing Unit	GPU
Gravity Search Algorithm	GSA
Gray Level Co-Occurrence Matrix	GLCM

TABLE 1. *(Continued.)* Glossary table.

Gray Wolf Optimizer	GWO
Green Histogram Channel Equalization-based Filtering	GHCEAF
Hard Exudates	HE
Hemorrhages	HM
Histogram Equalization	HEQ
Hybrid Genetic and ant Colony Optimization	HGACO
Hybrid Inductive Machine Learning Algorithm	HIMLA
Hyperparameter Tuning Inception-v4	HPTI-v4
Improved Contoured Convolutional Transformer	IC2T
Improved Coordination Attention Mechanism	ICAM
Indian DR Image Dataset	IDRiD
Intersection Over Union	IoU
k-nearest neighbors	KNN
Logistic	Lg
Logistic Model Tree	LMT
Logistic Regression	LR
Machine Learning	ML
Magnetoencephalography	MEG
Mean Absolute Error	MAE
Mean Squared Error	MSE
Microaneurysms	MA
ML-Based Neovascularization Detection	MLBNVD
mobile Health	mHealth
Modified Expectation Maximization	MEM
Modified Fuzzy Clustering Method	MdFCM
Multi Layer Perceptron	MLP
Multiclass DR Severity Classification	MCDR
Multilayer Perceptron	MLP
Multiple Instance Learning	MIL
Multi-Class SVMs	MSVMs
Network-in-Network	NiN
Nonproliferative DR	NPDR
Optic Disc	OD
Optical Coherence Tomography	OCT
Optical Coherence Tomography Angiography	OCTA
Optical Discs	ODs
Parallel Convolutional Neural Network	PCNN
Principal Component Analysis	PCA
Principal Component Analysis	PCA
Proliferative DR	PDR
Random Forest	RF
Retinal Fundus Image	RFI
Retinal Vascular Network	RVN
Root Mean Squared Error	RMSE
Run Length Matrix	RLM
Spatial Attention-U-Net	SA-U-Net
Sensitivity	SEN
Sequential Minimum Optimization	SMO
Simple Logistic	SLg
Soft Exudates	SE
Spatial Pyramid Pooling	SPP
State-of-the-art	SOTA
Superficial Capillary Plexus	SCP
Superficial Vascular Plexus	SVP
Support Vector Machine	SVM
True Positive Rate	TPR
Two Dimensional	2D
University of California San Diego	UCSD
Vision Transformer	VT
War Strategy Optimization	WSO

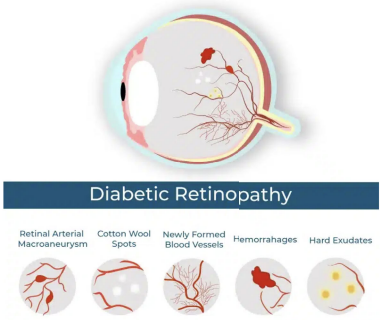


FIGURE 1. DR retinal cross-section image [7].

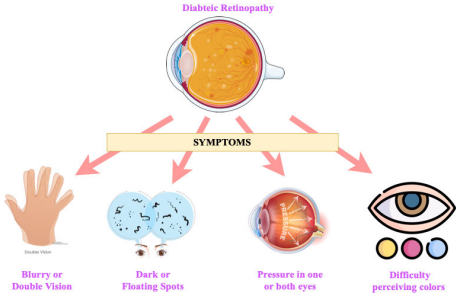


FIGURE 2. DR symptoms.

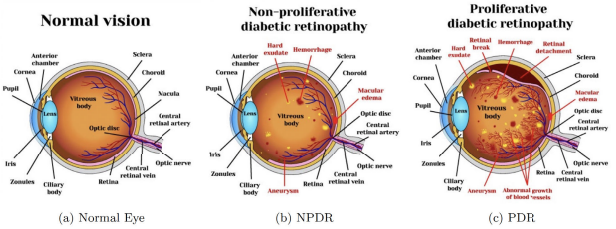


FIGURE 3. Stages of DR [11].

therapy [2]. Routine eye screening is important for knowing the sight-threatening stages of the disease and treating proliferative DR (PDR) and DME before irreversible vision loss occurs. A cost-effective method that has been employed in many European and non-European countries is eye screening. When combined with improved diabetes control, this has dramatically lowered the risk of vision loss [3]. However, the burden on healthcare systems increases with the number of diabetes patients, making automated solutions necessary to improve diagnostic accuracy and expedite the screening process.

*The Abbreviations and their respective full forms of all the terms used in this article can be found in Table 1.

DR is an ocular condition that gradually emerges due to diabetes, and it tends as the primary reason behind retinal vision impairment. This condition leads to the leakage and blockage of small blood vessels in the retina. DR often remains inconspicuous until patients encounter vision deterioration, making it a latent ailment. The prevalence of DR has surged in tandem with the longer life expectancy of diabetes patients. Furthermore, DR is the most concerning complication arising from diabetes, elevating the risk of developing additional conditions like kidney problems, heart ailments, and even mortality [4], [5]. The emergence and advancement of DR are primarily linked to three crucial factors: inadequate management of blood sugar levels, elevated blood pressure, and extended duration of diabetes [6].

As shown in Figure 1, Macroaneurysms (MA), cotton wool spots, hemmorahages (HM), hard exudates (HE) and Irregular blood vessel growth within the retina is a potential outcome of DR, leading to retinal scarring or bleeding, ultimately culminating in blindness. This progression can result in gradual vision loss, potentially leading to complete blindness in advanced stages. On a global scale, DR accounts for 2.6% of blindness cases [8]. The key risk factors most closely linked to the development of DR include the duration of diabetes, elevated hemoglobin A1C levels, and high blood pressure readings [9]. Regular screenings are essential for individuals with diabetes to ensure early detection of DR.

DR does not have any early symptoms but the symptoms are developed at the later stages. Figure 2 depicts the serious symptoms caused due to DR as follows:Blurred vision, fluctuating vision, difficulty with color perception, floaters (spots or dark strings floating in vision), dark or empty areas in vision, eye vision loss [10]. DR can also be divided into two primary stages, namely PDR Figure 3(a) and Non-proliferative DR (NPDR) Figure 3(b). Damage to the blood vessels in the retina can result in the leakage of fluid and proteins into the retinal tissue, leading to swelling. At this stage, it can be observed that the presence of HM, MA, and, is collectively referred to as NPDR. The absence of neovascularization determines the diagnosis of NPDR, although it may still incorporate typical diabetic retinopathy lesions. DR advances progressively, increasing in severity from mild to moderate and eventually reaching the potentially vision-threatening severe PDR stage. Accurate severity classification of the DR assists in identifying high-risk patients, which in turn aids in addressing potential issues through appropriate referrals, periodic checkups, and suitable treatments to maintain the current level of vision [12]. PDR represents the later stages of DR and signifies an angiogenic response in the retina. Angiogenesis is a natural process in which new blood vessels develop from existing ones [13]. Neovascularization in the retina often appears as the proliferation of new blood vessels along what is known as vascular arcades within the retina [14].

Thus, the diagnosis of DR involves collecting patient information through clinical history and interviews, conducting physical examinations, performing diagnostic tests, and seeking advice from other healthcare professionals. These methods are crucial for gaining insights into the patient's health condition. However, these traditional methods can be time-consuming, resource-intensive, and may not always be readily accessible, particularly in remote or underserved areas. Automated solutions, such as those based on Machine

TABLE 2. Review articles comparison Classification (CL), Grading (GD), Diagnosis (DG), Segmentation (SG), Detection (DT).

Reference Paper	Applications of AI in DR					Dataset Available	Case Studies	Methods		
	CL	**GD**	**DG**	**SG**	**DT**			**ML**	**DL**	**XAI**
[15]	✓	✓	✓	✓	✓	✗	✗	✗	✗	✗
[16]	✗	✗	✗	✓	✗	✓	✗	✓	✓	✗
[17]	✓	✗	✗	✗	✗	✓	✗	✓	✓	✗
[18]	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗
[19]	✗	✗	✗	✗	✗	✓	✗	✓	✓	✗
Present Work	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Learning (ML) and Deep Learning (DL), offer significant advantages in addressing these challenges. ML algorithms can analyze large volumes of retinal images (RI) swiftly and accurately, identifying signs of DR with high precision. DL, a subset of ML, excels in recognizing complex patterns and features in medical images, often surpassing human diagnostic capabilities. By leveraging these technologies, automated systems can provide rapid, consistent, and accurate screening for DR, Facilitating early detection and prompt intervention. This not only enhances patient outcomes but also alleviates the workload on healthcare professionals and systems, making quality eye care more accessible to a broader population.

Retinal imaging techniques include Fundus images (FI), Optical Coherence Tomography (OCT) images, and Optical Coherence Tomography Angiography (OCTA) images. Fundus imaging primarily focuses on the anatomical structures of retinal blood vessels, allowing for the observation of structural changes in these vessels. Conventionally, Fundus imaging was the main method for detecting retinal disorders. However, OCT has emerged as a widely adopted, non-invasive imaging technology that provides both quantitative and qualitative assessments of retinal structure in real-time. Its sensitivity to minor changes in retinal layers has made it particularly effective for diagnosing retinal issues [20], [21], [22]. Additionally, OCTA is a non-invasive technique that offers a three-dimensional view of the perfused blood vessels in the retina and choroid [23], [24]. Unlike standard OCT, which primarily analyzes the intensity of reflected light, OCTA examines both the intensity and the temporal variations of the OCT signal [25]. Existing literature on DR has extensively explored the potential of ML and DL in revolutionizing the diagnosis and classification of DR. Numerous studies have demonstrated efficacy of ML and DL algorithms in analyzing RI to detect DR with promising accuracy. For instance, convolutional neural networks (CNN), a type of DL architecture, have shown remarkable proficiency in identifying various stages of DR by learning intricate features from large datasets of annotated RI. Although there exist some literature surveys in the field of DR, this survey aims to provide a detailed overview of the latest advancements in AI, methodological approaches, and practical applications of ML and DL in DR. Thus, the motivation for writing this current survey paper stems from exploring the different applications of DR utilizing AI methodologies, such as classification, segmentation, early detection, early diagnosis, and grading.

This section presents some recent research on AI-based DR detection [15], [16], [17], [18], [19] whose characteristics are shown in the Table 2. An in-depth qualitative assessment of AI for DR is done by [15] in their review to scrutinize the state-of-the-art methods. The review explores segmentation methods for DR detection, including techniques for identifying MA, HM, and EX. A systematic literature review (SLR) is done by [16], which thoroughly investigates the foundational aspects of DR, including its underlying causes, the difficulties ophthalmologists face in detecting it, and proposed methods for early detection. Furthermore, the review offers a thorough assessment of modern AI and ML techniques and system models used in the diagnosis of DR. Reference [17] in their research paper, provided a consolidated and more comprehensive review, covering the aspects of dataset coverage, binary classification, Multiclass classification, ML based approaches, DL based approaches, Diabetic Retinopathy Classification (DRC) tools, TL experimentation and have highlighted the recent developments in automated-based DR classification techniques in ML and DL. Reference [18] have shed light on the selection and characteristics of eligible studies, quality assessment, threshold analysis, and heterogeneity test, meta-regression and sensitivity analysis, and publication bias. The objective is to methodically evaluate the diagnostic efficacy of an AI algorithm model for various forms of DR based on prospective studies conducted within the past five years. Furthermore, the review aims to explore

the factors that impact the accuracy of the algorithm in diagnosis. Reference [19] have investigated how AI is utilized in screening for DR through retrospective, prospective, and real-world studies, with a specific focus on analyzing AI's effectiveness using color retinal photographs. Moreover, the review discussed important governance concerns and obstacles related to implementing AI for DR screening in clinical practice, providing valuable perspectives for potential adopters. It is an in-depth study on AI algorithms for DR, publicly available datasets for DR, retrospective validation studies from ML, prospective validation studies of DL, AI analysis of smartphone photographs, different types of retinal photographs, different manufacturers of conventional cameras, mydriasis or non-mydriasis. Table 2 shows the comparison of this survey with the other existing surveys such as applications of AI in DR, dataset used, case studies, and methods used.

The main contribution of this article can be summarised as follows:

- Highlighted the standard benchmarking datasets and performance metrics available for conducting research in DR.
- A focused discussion on different applications of DR leveraging the AI approaches.
- Presented a few significant case studies utilizing AI techniques in real-life problem solving.
- An engaging discussion on the adoption of Explainable Artificial Intelligence (XAI) techniques in the field of DR.

The comprehensive review of existing survey papers highlights valuable insights from previous research, yet our current survey aims to address notable gaps in the existing literature. The review articles considered for the literature review are deficient in the application-specific detailed analysis of AI in DR and we incorporate the same in our review article along with a special focus on the benchmark datasets, and case studies. This paper is structured into several sections. Section II delves into the practical applications of AI in DR and highlights notable advancements in DR utilizing ML, DL. In Section III, case studies are presented. Section IV presents the XAI for DR, Section V outlines the challenges and Future Scope inherent in the current system. Finally, Section VI offers the conclusion.

A. METHODOLOGY

This review aims for exploring application of different imaging methods for DR classification, segmentation, detection, diagnosis. The review carried out is done based on the criteria which can be seen in Table 3. The articles from various journals such as Elsevier, Springer, IEEE, Taylor & Francis, Frontiers, Nature, MDPI satisfying the criteria mentioned in Table 3 alone are considered for the review and filtration is done accordingly. Figure 4 presents the article distribution across various Journals. Figure 5 presents the temporal distribution of articles for the years 2018-2024.

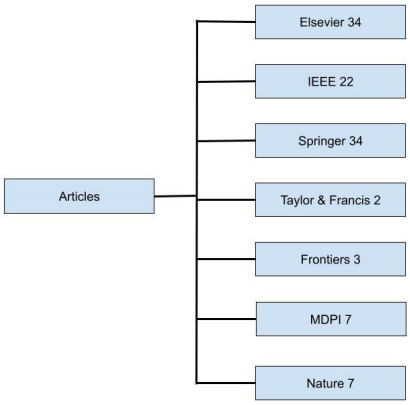


FIGURE 4. Distribution of article across the journals.

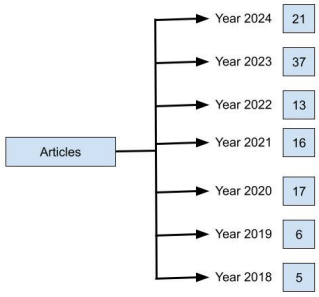


FIGURE 5. Temporal distribution of article.

Figure 6 presents the flowchart of analyzed and excluded articles considered for the review.

II. APPLICATION OF AI IN DR

Artificial Intelligence systems employing digital fundus photography devices have been developed to aid DR screening, to mitigate the increasing need for screening resulting from the expanding diabetic population across the globe [26]. This section illustrates the implementation of AI in DR using the ML and DL approaches. Figure 7 illustrates the distribution of existing research works conducted for different DR applications such as DR classification, early diagnosis, segmentation, early detection, and grading.

Figure 8 shows the AI pipeline diagram representing the techniques used at each phase of the pipeline inspired by the diagram from [17].

AI has opened several new avenues in DR prediction and detection not just limited to screening but also spread across various applications. This technology not only alleviates clinical workload but also enables prompt interventions, ultimately improving patient outcomes, prognosis, and management. In recent years, numerous research studies have been conducted in DR using AI techniques. This section presents existing literature works utilizing these AI methodologies for DR classification, diagnosis, segmentation, detection, and grading.

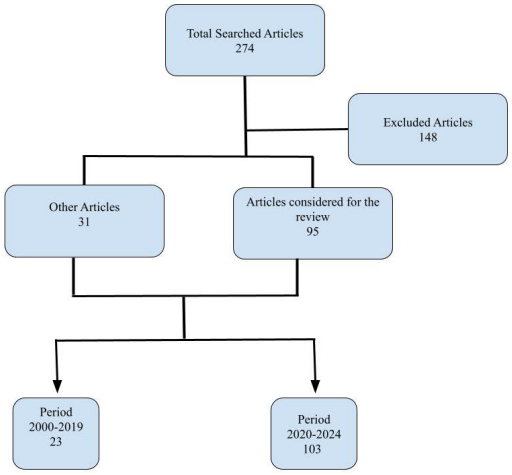


FIGURE 6. Flowchart of analyzed and excluded articles.

TABLE 3. Criteria for article inclusion and exclusion.

Inclusion Criterion	
I1	Scopus Indexed
I2	From Renowned Publisher (Elsevier, Springer, IEEE, Taylor & Francis, Frontiers, Nature, MDPI)
I3	Automated / AI powered Solutions for DR
I4	Articles related to FI, DL, XAI
Exclusion Criterion	
E1	Books
E2	Pre-Prints
E3	Magazines

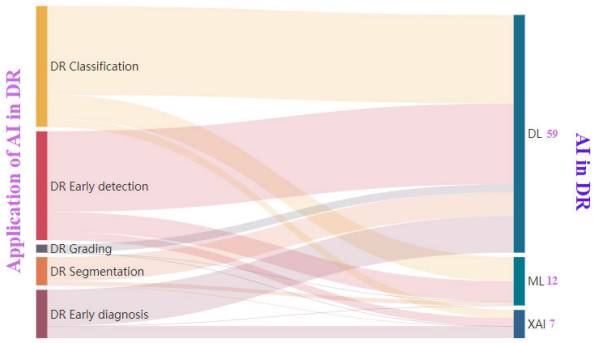


FIGURE 7. Distribution of AI in DR.

A. ML IN DIABETIC RETINOPATHY

The utilization of ML in the healthcare sector is unavoidable given its capability to aid in disease detection and management. Employing ML techniques for disease diagnosis can expedite decision-making processes and lower the occurrence of false positives. Figure 9 shows the usage of a specific ML model for DR in the literature considered for the study.

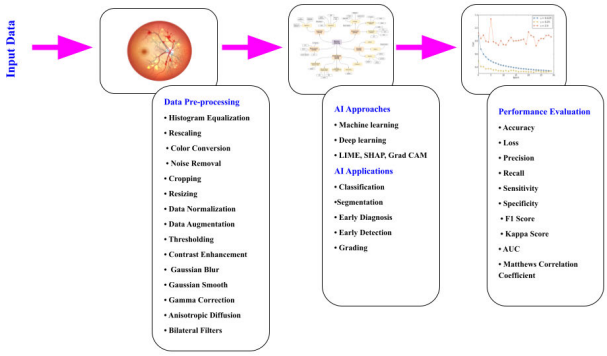


FIGURE 8. AI pipeline for DR applications.

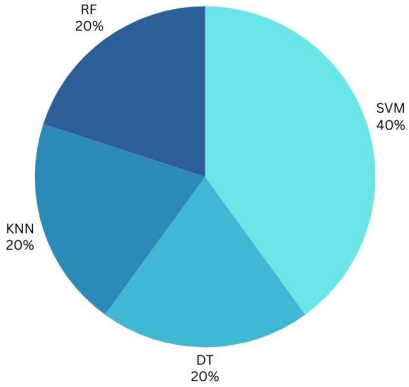


FIGURE 9. ML models distribution in DR.

Early detection of DR is aided by ML with the analysis of RI. Hybrid models are put at work that combine supervised learning with various techniques, enhancing accuracy even with inadequate datasets. Algorithms are trained on large datasets of annotated images to identify specific markers of DR, such as MA and HE, with high accuracy. This enables timely intervention and treatment, potentially preventing vision loss in affected patients. Classification techniques classify RI into stages of DR or detect the presence of the disease, aiding in early diagnosis and treatment. SVM [27], [28], [29], [30], [31], [32], [33], and RF [30], [34], [35], [36], [37], [38] are popular ML models for DR, known for their robust performance in classifying RI and aiding in early detection and management of the disease. The major challenges include dataset biases and complexities in grading due to the presence of multiple features, posing difficulties in accurate classification and diagnosis using ML techniques. However, The next viable approach to mitigate data bias is to enhance the training data. leveraging DL for its ability to extract complex features from RI, enhancing accuracy in disease detection and classification. By integrating DL with traditional algorithms, these models overcome data limitations, improving diagnostic performance for timely DR detection and treatment. The research that has been done so far on the early detection of DR using ML techniques is presented in this section.

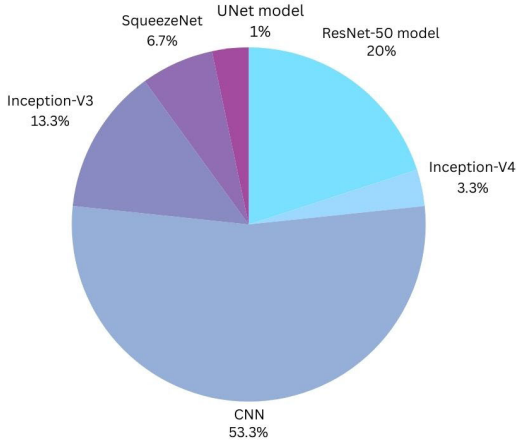


FIGURE 10. Distribution of variants of CNN models in DR.

B. DL IN DIABETIC RETINOPATHY

DL is a branch of AI designed to mimic the functioning of the human brain. Its primary aim is to enable computers to perform tasks that are typically intuitive for humans. DL finds widespread application in organizing unstructured or unlabeled data and discerning patterns within it. In the medical field, DL has had a profound impact, particularly in areas such as medical imaging, disease diagnosis, Medical Drug Discovery, Genome Synthesis, and more. The advancement of DL in this sector has been remarkable, largely driven by advancements in data processing techniques and the types of data utilized in models. The nature of data, whether preexisting or curated, can significantly impact the efficacy of a DL model. DL is increasingly utilized in diagnosing a variety of diseases, spanning from skin lesions and brain disorders to chronic illnesses. Figure 10 shows the usage of a specific DL model for DR from the literature considered for the study. All the DL models are from the family of CNN, the models with specific architecture details are provided with the respective percentage of usage and the unnamed CNN architectures are represented as a whole. DL is highly effective in diagnosing DR by analyzing RI with precision, often outperforming human experts. Trained on extensive datasets, these models detect subtle disease indicators automatically. Early detection facilitated by DL improves patient outcomes through timely intervention. Ongoing research aims to enhance accuracy and reliability, advancing its role in DR diagnosis. Among all the works referred to for the literature survey, 95% of the works are done on CNN. A superior performance is achieved in detecting DR by accurately identifying early signs in RI through automated processes. But several hurdles are encountered during the classification such as dataset quality, interpretability, and generalization across various populations and clinical environments for DR. Addressing these issues is critical for optimizing DL's role in improving diagnostic precision and patient outcomes in DR. AI can be applied effectively at various stages of DR for classification, segmentation, early detection, diagnosis, and grading.

C. AVAILABLE DATASET FOR DR APPLICATIONS

Table 4 presents the datasets categorized by their names, dataset sizes, distribution of images, types of cameras used for image collection, and applications. The review of various articles relevant to DR highlighted the most widely used high quality datasets utilized in DR prediction and detection such as OCTA [39], DRIVE [40], EyePACS [41], APTOS Kaggle database [42], STARE [43], IDRiD [44], MESSIDOR [45], DIARETDB1 [46], CHASE [47], HRF [48], Ocular Disease Intelligent Recognition (ODIR) [49], Diabetic Retinopathy Debrecen dataset from UCI ML repository [50]. Figure 11 shows the distribution of various benchmarking datasets used in existing research works we have included in our literature study. The most widely used datasets are MESSIDOR, IDRiD, DIARETDB, and EyePACS. Table 5 presents the preprocessing techniques used by various works considered for the study over benchmark datasets. It captures the diverse range of techniques employed in processing OCTA and RI, highlighting the importance of tailored methods for effective analysis and interpretation.

The image processing techniques used in OCTA and various retinal imaging datasets encompass a diverse range of methods aimed at enhancing image quality and extracting meaningful features. Gabor-based filters and Gaussian filtering are commonly employed for overall image enhancement and noise reduction, while wavelet transforms and new radial invariant moments (QRHMI) aid in feature extraction. Contrast enhancement is frequently achieved through Contrast Limited Adaptive Histogram Equalization (CLAHE) and histogram equalization (HE). Noise and artifact removal techniques, including adaptive filtering, are essential for improving image clarity, particularly in datasets like DRIVE and APTOS. Morphological operations, such as contour detection, enhance the visibility of blood vessels. Advanced methods like the Artificial Bee Colony (ABC) technique and Multiscale Retinex with Chromaticity Preservation (MSRCP) further optimize image analysis. Image resizing and transformations, including cropping and rotation, standardize images for consistent analysis. Specific datasets employ tailored approaches: DRIVE focuses on noise elimination and green channel extraction; EyePACS uses Tophat transformations for background correction; and APTOS emphasizes noise and artifact removal. Other datasets like STARE, IDRiD, MESSIDOR, and DIARETDB1 utilize unique combinations of filtering and normalization techniques to enhance image quality, demonstrating the importance of customized methods for effective analysis and interpretation in retinal imaging.

D. DR CLASSIFICATION

AI-based data classification methods help AI systems to sort data into predefined categories or labels. By examining historical data patterns, AI classification methods effectively organize extensive datasets into meaningful patterns or features used in DR classification, particularly for diagnosing disease symptoms. The following subsection discuss the

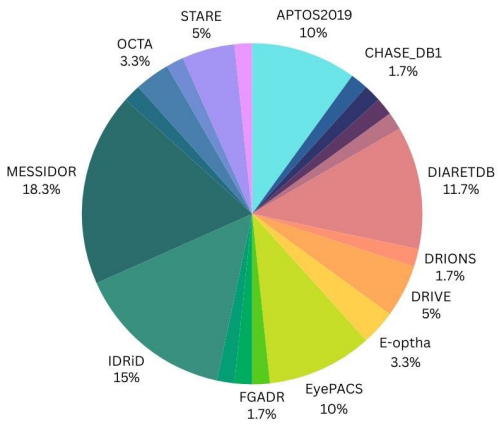


FIGURE 11. Benchmark dataset distribution in DR.

influence of DL and ML in DR classification. Table 6 summarizes existing literature employing ML and DL techniques for DR classification.

1) ML FOR CLASSIFICATION

An enhanced DeepForest technique named Multi-class feature extraction deep forest (MFgcForest) method is designed by Xiwen et al. specifically for the multi-classification problem of DR [51]. The designed method involves feeding unprocessed data into the MFgcForest algorithm, where sub-samples taken as input using multi-grain scans are initially classified by both RFs to determine the patient's health status. Following this, certain low-level features are eliminated based on their classification performance to prevent any adverse impact upon the final outcome. The filtered features that are left are then fed into a cascade forest to produce the final prediction results, which improves the classification performance. The experimental study utilized the kaggle diabetic retinal image dataset. In comparison to other ML algorithms like KNN, SVM, and RF, the designed algorithm, called MFgcForest, showed superior performance. It achieved an accuracy of 74%, a recall rate of 74%, a precision of 63%, and an F1 value of 64%. This improvement in predictive classification accuracy of diabetic data by 2-4% holds considerable theoretical and practical value for diabetes diagnosis. In future developments of the algorithm, emphasis could be placed on minimizing the model's parameters.

Khalili Pour et al. [29] have conducted a study to distinguish between eyes affected with NPDR and PDR through the analysis of OCTA vascular density maps. The Farabi Eye Hospital at Tehran University of Medical Sciences, Iran provided the RI dataset comprising 78 DR patients of which 103 are NPDR and 45 are PDR patients. Initially, the researchers applied Gabor-based filters to enhance the image quality. They then classified these images into two categories: NPDR and PDR. To differentiate between PDR and NPDR images, the authors utilized a supervised ML technique called the genetic evolutionary algorithm and an optimized SVM classifier. The study also utilized OCTA vascular density maps at two levels: deep capillary plexus (DCP) and superficial capillary plexus (SCP). The research findings indicate that the vascular density map of the deep retinal layer performed exceptionally well, achieving a remarkable accuracy of 90% in differentiating between NPDR and PDR.

ML techniques were used on low-resource datasets by Sengupta et al. [30] for classification. The computation is done using a publicly available dataset Antal DB with only 19 attributes describing 1151 occurrences. The authors used the eight most popular ML techniques for classification such as Alternating DT, KNN, AdaBoost, MLP, Naïve Bayes, RF, SVM, and Pattern Classifier and the Logistic Regression(LR) approach yield the best results, with an accuracy of 77.8% and F1-Score of 0.712. The authors utilized the eight most popular ML techniques for classification, including alternating KNN, DT, MLP, Naïve Bayes, RF, AdaBoost, SVM, and Pattern Classifier. The LR approach produced the best results, achieving an accuracy of 77.8% and an F1-Score of 71.2%. Further stacking all ML algorithms improved the accuracy to 81.3% and the F1-Score to 78.4% Additionally, the confusion matrix revealed that only approximately 12% of erroneous negative values (37 points out of 306 cases) were identified. In the study's extension, the authors recommend incorporating a graded classification of DR into four phases and using the same dataset to determine the presence of macular edema.

Liu et al. [31] a model was developed to differentiate between individuals with healthy controls (HC) and patients with DR using OCTA images measuring 3 × 3 mm and various ML models. The OCTA angiography dataset consisted of images from the deep vascular plexus (DVP), superficial vascular plexus (SVP), and retinal vascular network (RVN). The dataset included 19 DR patients (38 eyes) and 25 HC individuals (50 eyes). The collected dataset was pre-processed using wavelet transform to extract features from RFI. Four ML models, including elastic net-regularized LR (LR-EN), LR, XGBoost gradient boosting tree, and SVM were used to classify the wavelet features between the two groups. LR-EN and LR, using the entire image set, showed a specificity of 80% and the highest sensitivity of 84%, resulting in the best diagnostic accuracy of 82%, with AUC values of 83% and 0.84%, respectively. These values were slightly lower than those achieved by LR using a single image (SVP) of 85% or a combination of two images (DVP and SVP) of 85%. The proposed model has limitations, including a relatively small sample size for each cohort, consideration of OCTA images only from mild DR patients, and the exclusion of RI, intra-retinal layers, and retinal angiography thickness maps.

Geetha et al. [35] proposed a DL approach for precise vessel segmentation using 2D Gabor Wavelets and matrix convolution. They used publicly accessible datasets such

[1]NA-Not Applicable.
[2]RI-Retinal Image.

TABLE 4. Benchmarking dataset analysis.

Dataset	Size	Image Distribution	Capturing Device	Application
OCTA angiography dataset [39]	19 DR patients, 38 RI and 25 HC patients, 44 RI	DR: 114 and HC: 132 images	Optovue OCTA device	Classification
DRIVE [40]	40 RI	Disease: 7, no disease: 33	Canon CR5 nonmydriatic 3CCD camera	Classification & Segmentation
EyePACS [41]	35,126 RI	Mild: 2443, Moderate: 5292, Severe: 873, PDR:708, No-DR: 25810	Canon CR-DGi and CR-1 nonmydriatic cameras	Classification, Segmentation, Early Detection & Early Diagnosis
APTOS Kaggle database [42]	3662 RI	Mild: 370, Moderate: 999, Severe: 193, PDR: 295, No-DR:1805	Retinal fundus camera	Classification, Early Detection & Early Diagnosis
STARE [43]	20 RI	Disease: 10, no disease: 10	Topcon TRV-50 fundus camera	Classification & Segmentation
IDRiD [44]	516 RI	Training mild: 20, moderate: 136, severe: 74, PDR: 49, No-DR: 134. Testing mild: 5, moderate: 32, severe: 19, PDR: 13, No-DR: 34.	Kowa VX-10	Classification, Segmentation, Early Detection & Early Diagnosis
MESSIDOR [45]	1200 RI	Mild: 153, Moderate: 247, Severe: 254, No-DR:546	Topcon TRCNW6 non-mydriatic fundus camera	Classification, Segmentation, Early Detection & Early Diagnosis
DIARETDB1 [46]	89 RI	MA: 80, SE:36, HE: 48, HM: 54, No-abnormality: 5	Digital fundus camera	Classification, Segmentation, Early Detection & Early Diagnosis
CHASE [47]	28 RI		Nidek NM-200-D	Classification, Segmentation, Early Detection
HRF [48]	45 RI	DR: 15, glaucoma: 15, No disease: 15	Canon CR-1 fundus camera	Classification, Segmentation
Ocular Disease Intelligent Recognition (ODIR) [49]	5000 RI	Total 4784	Canon, Kowa VX-10, Zeiss	Early Detection
DR Debrecen dataset from UCI ML repository [50]	1151 RI	Instances 1151	NA	Early Detection

TABLE 4. *(Continued.)* **Benchmarking dataset analysis.**

Large Dataset of Labeled OCT and Chest X-Ray Images[58]	108,312 RI	37,206 with choroidal neovascularization, 11,349 with diabetic macular edema, 8,617 with drusen, and 51,140 normal	NA	Early Diagnosis
OCTID: Optical Coherence Tomography Image Database[59]	More than 500 high resolution RI	Image classes include Normal (NO), Macular Hole (MH), Age-related Macular Degeneration (AMD), Central Serous Retinopathy (CSR), and DR	NA [1]	Early Diagnosis
OCTDL: Optical Coherence Tomography Dataset for Image-Based DL Methods[60]	Over 2000 RI[2]	AMD - 1231 images; DME - 147 images; Epiretinal Membrane- 155 images; Normal - 332 images; Retinal Artery Occlusion - 22 images; Retinal Vein Occlusion - 101 images; Vitreomacular Interface Disease - 76 images.	Optovue Avanti	Early Detection

TABLE 5. **Pre-processing techniques applied over benchmarking dataset.**

Dataset	Pre-processing Technique Applied
OCTA vascular density maps	Gabor-based filters to enhance the image quality [29], wavelet transform to extract features from RFI [31]
DRIVE	Noise elimination, followed by green channel extraction and enhancement using CLAHE and gamma correction [61], normalization and noise removal techniques are applied and later an adaptive filtering based on green histogram channel equalization is employed for analyzing blood vessels in images [53], gaussian filtering and adaptive histogram equalization [35], enhancing contrast, applying thresholding, removing backgrounds, extracting the green channel, detecting contours, and performing morphological operations [62]
EyePACS	Image resizing, green channel extraction, Tophat Transformation (White Tophat), and BottomHAT Transformations (Black Tophat) were used [37], noise removal and contrast enhancement of the input images [32], The artificial bee colony (ABC) technique [63], shape cropping, cropping, and Gaussian blurring [64]
APTOS	Noise removal, artifact removal, CLAHE, and image resizing [65], new radial invariant moments of Hahn in quaternion (QRHMI) was employed [66], Gaussian filter for coherence, and image transformations like resizing, rotation, and color adjustments [34], green channel, CLAHE [33], CLAHE [67], [68]
STARE	Noise reduction and local contrast enhancement [69], CLAHE [54]
IDRiD	Intensity normalization and HE [70], data normalization, histogram equalization, and proportional resizing [71], RoI and median filtering extraction to eliminate noise [72], optimal bandpass filter method [73]
MESSIDOR	CLAHE and Gaussian filtering [74], image resizing, noisy image removal [75],blurring, grayscale conversion, and mask application to remove unnecessary dark areas [76], CLAHE [77], filtering and green channel extraction techniques [27]Gabor filtering (GF) for noise reduction [78], CLAHE [79], CLAHE and HE [80]Multiscale Retinex with Chromaticity Preservation (MSRCP) for improving image contrast [81], noise channel and median filters, CLAHE [82]
DIARETDB1	Mix of mathematical morphology, Gaussian match filtering, and entropy thresholding [83], normalization [84], feature scaling, masking, and resizing [36]
CHASE	Normalization [85]

as DRIVE, STARE, DIARETDB, and Kaggle DR for model training. The datasets were preprocessed using image preprocessing techniques like gaussian filtering and adaptive histogram equalization. The Gini-index was used to extract

a total of 36 relevant features from the vascular map. An RF classifier was used to effectively classify RI into healthy, NVE, and NVD categories. The efficacy of the method was evaluated on widely accessible databases, and the accuracy and Out-of-bag (OOB) errors were used as evaluation metrics. Comparative analysis with Online Sequential Extreme Learning Machine (OS-ELM), Quadratic SVM (Q-SVM), and Mediods classifiers showed that the RF classifier outperformed the others. It achieved an average accuracy of 98% with an OOB error below 0.05. However, there were cases where the algorithm mistakenly classified NVD images as NVE due to the presence of other pathological factors such as MA or EX. Future research should explore implementing DL networks to achieve improved results with reduced computational complexity.

The efficacy of ML methods in segmenting and classifying DR is explored by Ali et al. [85]. The two-dimensional (2D) RFI encompassing various DR severity levels such as moderate, mild, proliferative, non-proliferative, and normal were sourced from 500 patients at Bahawal Victoria Hospital (BVH)in Bahawalpur, Pakistan. The acquired RFI were translated to gray scale format and normalization was performed using HE. The authors proposed a new automatic region-growing segmentation framework based on clustering. Five ML classifiers, namely logistic model tree (LMT), sequential minimum optimization (SMO), logistic (Lg), and simple logistic (SLg), multi-layer perceptron (MLP), were deployed using 10-fold cross-validation. The results obtained were 77.67%, 89.87%, and 96.33% 80%, accuracy, respectively. Notably, the SLg classifier outperformed all other classifiers and achieved an accuracy rate of 99.73% overall. These findings highlight the limitations of the supervised learning-based classifier when applied to the retina FI dataset from the ophthalmology department at the BVH, Bahawalpur, Pakistan.

2) DL FOR CLASSIFICATION

DR is a common eye condition caused by diabetes mellitus (DM) and continues to be a major cause of blindness globally. Karthika et al. [86] discussed DR classification by training and testing a transformer network by utilizing publicly available FI datasets from Kaggle (EyePACS) in conjunction with real-time datasets. The proposed model aimed to automate DR classification at earlier stages. Key features such as blood vessels, HM, SE, and HE were segmented from preprocessed images using the D2-U-Net technique to enhance classification accuracy. Additionally, a novel SE-ResCA-GTNet approach was introduced to distinguish varying severities of retinopathy-affected patients. The Gazelle Optimization (GO) algorithm was employed to fine-tune the proposed classifier, resulting in improved accuracy. Using the FI dataset, the SE-ResCA-GTNet classification model achieved high, recall, and F1 measures, with an accuracy of 99.4%, 99.3%, and 99.8% respectively. Despite the promising results, the study faced limitations such as the need for challenges in handling varying image qualities, comprehensive datasets, longer training times, dataset bias, and the necessity for clinical validation. Future research directions may include dataset augmentation, domain adaptation, and effort toward explainability and interpretability in the model.

Detecting DR automatically is crucial due to its significant contribution to irreversible vision loss, particularly among economically active populations in developed countries. This work of Sunkari et al. [87] focuses on DR detection and analysis of its different stages using DL algorithms applied to FI. The model used real-time datasets from hospitals, with FI as the input. The datasets included the APTOS Kaggle database and the FI (SN Dataset) was collected from the Vision Research Foundation and Madras Diabetes Research Foundation in Chennai. The suggested ResNet-18 architecture, which incorporated a swish function, achieved impressive performance metrics. These included a sensitivity of 93.42%, precision of 93.77%, F1-score of 93.59%, and an accuracy of 93.51%. As a future extension, the development of a DR screening website based on this research is suggested.

Sivapriya et al. [60] propose a DL model for automating the segmentation of blood vessels and classification of DR using FI. The proposed system comprises three key stages: preprocessing, vessel segmentation, and classification. Initially, input images undergo noise elimination, followed by green channel extraction and enhancement using CLAHE and gamma correction. Addressing the crucial aspect of Retinal Vascular Structure (RVS) segmentation for detecting various DR stages, the U-Net architecture serves as the basis for the segmentation model. To ensure comprehensive semantic information retention, ResEAD2Net is introduced, modifying the downsampling and upsampling layers while incorporating residual blocks for enhanced computational speed. The proposed Regularized Random Walker (RRW) algorithm focuses on improving structural connectivity, specifically in relation to broken blood vessels. The vessel structure's features are extracted and then fed into an ML classifier to predict DR grading. The model is trained using the STARE, DRIVE, and MESSIDOR2 datasets. The proposed approach demonstrates superior performance in segmentation, achieving the following metrics for the specified datasets Accuracy, Specificity, Sensitivity, AUC values of 99.01%, 98.07%, 90.24%, 97.51% for the STARE dataset, 97.55%, 98.01%, 90.07%, 97.64% for the DRIVE dataset. In addition, the features obtained from the segmentation and ML classifier outperform current methods in multiclass classification, achieving an accuracy of 98.88%, a sensitivity of 98.91%, and a specificity of 98.29% on the MESSIDOR-2 dataset.

An automated examination to detect red lesions in RI has been developed by Saranya et al. [68] for early DR identification. Noise reduction and local contrast enhancement are the primary preprocessing steps for image segmentation. Red lesion semantic segmentation is achieved using the proposed U-Net architecture. CNN training and severity level classification are performed utilizing segmented images

TABLE 6. Summary of DR classification techniques.

Applications	Ref. Paper	Method	Dataset	Performance Metrics	Limitation
Classification	[86]	SE-ResCA-GTNet	EyePACS dataset, realtime datasets	Accuracy 99.4%	Key considerations encompass extensive datasets, prolonged training, dataset biases, managing diverse image qualities, and the critical need for clinical validation.
	[87]	ResNet-18	APTOS Kaggle database, The Fundusimages (SN Dataset) collected from Vision Research Foundation andMadras Diabetes Research Foundation in Chennai	Accuracy 93.51%	NA
	[61]	ResEAD2Net	STARE, DRIVE, MESSIDOR-2	Accuracy 99.01%	NA
	[29]	Genetic Evolutionary Algorithm, SVM	OCTA vascular density maps	Accuracy 90%	Excluded retinal edema, vitreous hemorrhage, the approach did not utilize ultrawide-field fluorescein angiography (UWF-FA)
	[69]	U-Net architecture	IDRiD, DIARETDB1, MESSIDOR, and STARE	Accuracy 96.65%	The proposed model often struggles with accurately detecting red lesions due to the intricate microstructure involved.
	[65]	RetNet-10 model	APTOS, Messidor2, and IDRiD	Accuracy 98.65%	The dataset is imbalanced primarily due to insufficient images in specific classes across the three fundus datasets.
	[88]	SqueezeNet	IDRiD	Accuracy 94.2%	The primary challenge in deploying blockchain technology for healthcare records is scalability.
	[73]	CapsNet	Messidor	Accuracy 98.64%	NA
	[70]	CNN model	MESSIDOR and IDRiD	Accuracy 96.85%	NA
	[65]	CNN-IHM	APTOS2019		Repeatedly applying the ACO during QRHMI parameter optimization causes time consumption issues.

TABLE 6. *(Continued.)* **Summary of DR classification techniques.**

[89]	FrWSO SqueezeNet	IDRiD	Accuracy 91.1%	NA
[70]	DTLCDS	IDRiD	Accuracy 99.36%	The study lacks a clinically acquired database, and adopts a trial-and-error approach for setting hyperparameters
[53]	CenterResnet-50	DRIVE, STARE, REVIEW, CHASE_DB1, HRF, and DRIONS fundus dataset	Accuracy 98%	NA
[30]	DT, KNN, AdaBoost, MLP, Naive Bayes, RF, SVM, LR	Antal DB A publicly available dataset with 19 features and 1151 instances that is accessible	Accuracy 77.8%	Excluding advanced techniques like DL, transfer learning, non-hyperparameter optimization, and graded classification approaches.
[90]	HGACO-based CNN model	MESSIDOR	AUC, specificity, and sensitivity rates of 95.78%, 94.56%, and 91.98% respectively	NA
[91]	GATL	APTOS 2019 and EyePACS	Accuracy 94.3%	NA
[74]	Multi-stream deep neural network	Messidor-2 (two categories) and EyePACS (two, five categories)	Accuracy 95.58%	NA
[75]	Deep extracting features from DenseNet models and EfficientNet, classification using RF and SVM algorithms	APTOS, Messidor-2	Accuracy 96.32%	NA
[31]	LR-EN, LR, XGBoost gradient boosting tree, and SVM	OCTA	Accuracy 82%	The study analyzed small cohorts of mild DR patients using only 3 X 3mm OCTA images, excluding RI, intra-retinal layers, and retinal angiography thickness maps.

TABLE 6. *(Continued.)* **Summary of DR classification techniques.**

	[35]	RF	Kaggle DR, DRIVE, STARE, and DIARETDB1	Accuracy 98%	The algorithm occasionally misidentifies NVD images as NVE due to additional pathological features like MA and EX.
	[92]	GCA-EfficientNet (GENet)	Kaggle competition dataset containing 35126 high-resolution color fundusimages	Accuracy 95.6%	NA
	[93]	CNN-based segment-level DRE-classifier	Kaggle and DIARET-DB1	AUCROC of 0.963, along with sensitivity and specificity of 96.37% and 96.37%	NA
	[85]	LMT, SMO, logistic, simple logistic, multi-layer perceptron	BVH RF Image Dataset & Pub-liclyAvail-ableDataset (Messidor-2 RF-Image)	Accuracy 99.73%	Feature selection using techniques such as F, MI, and PA on the BVH RF Image Dataset was constrained to selecting ex-actly 30 features.
	[76]	HPTI-v4, MLP	MESSIDOR	Accuracy 99.12%	NA
	[51]	MFgcForest	Kaggle diabetic retinal image dataset	Accuracy 74%	

obtained from the red lesion identification procedure. The experimental study was conducted using the four public datasets such as MESSIDOR, DIARETDB1, IDRiD, and STARE. The performance of the proposed approach is estimated as follows. Specificity, sensitivity, and accuracy attained for DR severity classification using the MESSI-DOR dataset were 93.8% 92.3%, and 94%, respectively. DIARETDB1 dataset obtained an sensitivity of 95.4%, an accuracy of 95.5%, and a specificity of 95.5%. When applied to the IDRiD dataset, the red lesion identification method achieved a sensitivity of 89%, accuracy of 95.65%, and specificity of 99%. For the STARE dataset, the obtained specificity, sensitivity, and accuracy are 96.65%, 95.23%, and 96%, respectively. Due to the complex micro-structure involved, the proposed U-Net architecture frequently had difficulty in accurately identifying red lesions. In addition, the diagnostic approach may be improved to identify DR in premature infants, newborns and children.

Usman et al. [94] developed a CNN-based DL model for classifying DR. The model is trained on the publicly accessible CSME dataset. Feature extraction is done with Principal Component Analysis (PCA). Following that, a pre-trained CNN architecture was used to introduce a DL-based Multi-Label Feature Extraction and Classification (ML-FEC) model. Three state-of-the-art CNN architectures ResNet152, ResNet50, and SqueezeNet1 were then trained on a selection of images using Transfer Learning (TL). The experimental findings showed that SqueezeNet1 achieved an accuracy of 91.94% with a Hamming loss of 0.0805, ResNet152 obtained an accuracy of 94.40% with a Hamming loss of 0.0560 and ResNet50 acquired a classification accuracy of 93.67% with a Hamming loss of 0.0603. The multi-label training is

limited by the lack of publicly available substantial training datasets with several recognized lesion labels. To enhance the accuracy of predictions in the field, additional studies on larger data sets can be carried out in the future.

Raiaan et al. [64] developed a DL algorithm that accurately classifies FI into five distinct DR groups: no DR (Grade 0), mild NPDR (Grade 1), moderate NPDR (Grade 2), severe NPDR (Grade 3), and proliferative DR (PDR) (Grade 4). The study utilized three diverse DR datasets (APTOS, IDRiD, and MESSIDOR2) resulting in a total of 5,819 initial images. Prior to model training, various image preprocessing techniques, including noise removal, artifact removal, CLAHE, and image resizing were employed, enhancing the overall quality of the image. Six state-of-the-art models such as Xception, VGG19, InceptionV3, ResNet50, MobileNetV2, and VGG16 are used for comparative study. Remarkably, the developed RetNet-10 model has a testing accuracy of 98.65%, outperforming all previous models. Testing accuracy is achieved by Xception, VGG19, InceptionV3, ResNet50, MobileNetV2, and VGG16 with respective results of 89.57%, 88.21%, 87.68%, 87.23%, 91.42% and 90.16%. Insufficient images for certain classes within the three fundus datasets are the primary cause of the imbalanced dataset. To address this, the dataset size must be increased. Additionally, various image processing techniques can be used in future for the better performance.

DR must be identified early to prevent individuals from blindness caused by untreated DM. A unique technique is presented by Uppamma and Bhattacharya [88] that includes an enhanced bio-inspired multi-domain feature extraction. In this approach, the authors have used adaptive thresholding to segment important elements such as the blood arteries, macula, optical disc, HM, and EX. The dataset used for the experimental study is the IDRiD dataset which is pre-processed using lesion segmentation and Median filtering techniques. The Taylor African Vulture Optimization (AVO) technique is used for tuning hyperparameters and extracting the most important features from the dataset using the firefly algorithm. The significant features are supplied to the SqueezeNet classifier for the prediction of the DR. The proposed approach has shown an accuracy rate of 94.2%. The primary challenge with implementing blockchain is scalability, especially when it comes to healthcare data. The Future work would be to focus on problems with the current setup, such as the extra expenses associated with incorporating blockchain technology into healthcare systems that are outdated even as third-party transaction fees are removed.

An improved capsule network CapsNet intended for DR identification and classification is developed by Kalyani et al. [73]. MESSIDOR dataset is considered for model training and FI of the dataset preprocessed using the techniques CLAHE and Gaussian filtering. To extract characteristics from FI, the main and convolutional capsule layers are used. The softmax layer class and capsule layer are used to determine the likelihood that the image belongs to a specific class. When dealing with healthy retinas, as well as stage 1 FIs, stage 2FIs, and stage 3 FIs, our capsule network achieves impressive accuracy rates of 98.64%, 97.98%, and 97.65% respectively. Due to complex lesion formations in FI, particularly in color images, identifying the 5 rigorous levels of DR might be a challenging task. Future work can be extended by training the suggested capsule network to cover all possible classes in DR.

A CNN model with a primary focus on early diagnosis for DR identification is developed by Ali et al. [69]. The study has used a unique end-to-end strategy for the extraction of features from FI by concatenating features from Resnet50 and Inceptionv3. The dataset used consists of OCT FI which is named DR detection from kaggle and experimented with several approaches to improve model performance, such as data augmentation and image enhancement. Advanced preprocessing methods such as intensity normalization and HE were applied to boost the model's sensitivity to DR indicators and enhance feature extraction. The model's performance is analyzed using significant measures that exceed state-of-the-art techniques such as specificity 98.92%, precision 96.46%, F1 score 98.65%, accuracy 96.85%, and sensitivity 99.28%.

Tahiri et al. [65] proposes a new hybrid model, CNN-IHM for DR classification by combining the discrete moment quadratic approach with ML and AI techniques. The developed hybrid model was evaluated through an experimental study using the APTOS dataset. To preprocess the APTOS dataset, the image preprocessing technique called new radial invariant moments of Hahn in quaternion (QRHMI) was employed. This technique is based on the fast and stable calculation of orthogonal discrete Hahn polynomials (HP), which are categorized by certain parameters. The ant colony meta-heuristic algorithm (ACO) was used to optimize the parameters of HP to achieve accurate classification. The results showed that the AUC for the five groups (moderate, mild, proliferative, severe, and healthy) were 97.02%, 97.83%, 94.87%, 94.58%, and 96.54% respectively, out of a total of 1795 images. Although the suggested method was shown to be effective, it is a time-consuming process because the ACO is applied repeatedly when optimizing the QRHMI settings.

DRC system has been developed based on the hybrid genetic and Ant colony optimization (HGACO) technique by Kukkar et al. [90]. The HGACO technique is employed along with the pre-trained CNN model RESnet. This technique aims to extract features from the ROI and categorize them into four levels of severity. To accomplish this, the DR images are first converted into red, green, and blue (RGB) channels. The ROI is then extracted using a K-means cluster-based growing region segmentation technique. Following this, the DR images are classified into four rigors categories using the HGACO-based CNN model. When comparing our developed DRC system to other state-of-the-art classification systems using the MESSIDOR dataset, our system obtained superior AUC, specificity, and sensitivity rates of 95.78%,

94.56%, and 91.98% respectively. Furthermore, patch-wise classification techniques can be applied to further enhance our developed model.

Beevi et al. [89] propose a multilayer severity classification model for DR classification. The model consists of two phases. In the first phase, the authors use FrWSO SqueezeNet for the first level of classification. This helps classify the FI into normal or pathological DR groups. To fine-tune the SqueezeNet model weights, they employ Fractional War Strategy Optimization (FrWSO), which combines War Strategy Optimization (WSO) with Fractional Calculus (FC). In the second phase, they use FrWSO Deep CNN (DCNN) to assess the severity level of abnormal images. To modify the weight of the DCNN, they apply the recommended Fractional War Royale Optimization (FrWRO) method, which combines FrWSO and Battle Royale Optimization (BRO). When tested on the Indian DR Image Dataset (IDRiD), the proposed model achieves the highest performance with an accuracy, specificity, and sensitivity of 91.1%, 91.3%, and 89.8%, respectively.

A Deep TL-based Computerized Diagnostic System (DTLCDS) for Multiclass DR severity Classification (MCDR) was developed by Vij et al. [70]. It modifies and compares the various deep inductive TL (DITL) models including ResNet34, VGG16, Inception V3, Xception, and EfficientNet B0. The primary goals of the proposed DTLCDS systems include preprocessing data labels which are inconsistent, introducing an improved DITL approach for feature extraction with a global average pooling layer to minimize loss through the use of leaky ReLU, and mitigating overfitting. For preprocessing, data normalization, histogram equalization, and proportional resizing techniques were introduced. The IDRiD dataset is used to automatically classify the severity stages of DR using a softmax layer for the final classification. Extensive analysis demonstrates that the proposed method surpasses existing approaches, achieving 98.6% F1-score, 99.7% AUC-ROC, 99.36% accuracy,98.6% recall,98.6% precision, and 99.02% AUC. Hence, the scope of the proposed system can be expanded by incorporating DL segmentation techniques to enable the detection of multiclass DR severity stages.

Malik et al. [53] proposed an ML-based classification model for detecting DR. The CenterResnet-50 classifies color FI using OCT for retinal imaging and an ensemble technique for vascular characterization. The validation set consists of 229 FI's from benchmarking databases (STARE, HRF, DRIONS, CHASE_DB1, DRIVE, and REVIEW) which are manually annotated by ophthalmologists guaranteeing the thorough assessment of the developed method. For image preprocessing, normalization and noise removal techniques are applied and later an adaptive filtering based on green histogram channel equalization is employed for analyzing blood vessels in images. The proposed method achieved 63% AUC, 67% sensitivity, 73% specificity, and 98% classification accuracy, highlighting how crucial regular eye examinations constitute for early DR diagnosis.

Hu et al. [91] introduced Graph Adversarial Transfer Learning (GATL) as a method for categorizing DR. GATL leverages adversarial training, graph neural networks, and self-supervised learning to improve model robustness and lower annotation costs. Initially, GATL employs self-supervised training to minimize annotation expenses in the target domain, distinguishing this method from supervised approaches by significantly reducing costs. Then, a graph neural network is incorporated to identify the latent features among unfamiliar samples. To bolster the model's resilience, adversarial training is utilized to align both intra-domain and inter-domain aspects and additionally boost the classification accuracy. GATL achieves a classification sensitivity of 97.5%, specificity of 91.1%, and accuracy of 94.3%, on the APTOS-2019 dataset. On the EyePACS dataset, it achieves an accuracy of 86.7%, sensitivity of 92.7%, and specificity of 95.7%. Because of their small size and similarity, GATL is particularly effective in identifying the presence of lesions. However, it encounters challenges when it comes to classifying fine lesions. Future studies should focus on complex patterns of classifying fine lesions in lesion categorization.

Mustafa et al. [74] developed a model using a multi stream ensemble deep network to categorize the rigors of DR. The authors developed a method that uses PCA and deep networks to extract changes within and across classes from the raw FI. PCA is used to reduce dimensionality, and ensemble classifiers (RF and AdaBoost) for better precision, DenseNet-121 and ResNet-50 for extracting features. The image processing techniques such as image resizing, noisy image removal, and image can be enhanced by implementing augmentation techniques. From pooling layers, the deep traits of the residual network and densely linked neural network are recovered. This study introduces a multi-stream deep neural network that utilizes 2, 3, and 5-category features to classify and grade DR using the MESSIDOR-2 and EyePACS datasets. The outcomes of the experiment demonstrate that the developed strategy performs better with an accuracy of 95.58% and is a feasible option for the automated diagnosis of DR.

A hybrid approach for classifying DR using feature selection via wrapper methods introduced by Canayaz et al. [75]. This approach utilizes image processing techniques, including blurring, grayscale conversion, and mask application to remove unnecessary dark areas, to enhance FI. It identifies the most impactful features by employing wrapper methods on 512 deep extracting features from DenseNet models and EfficientNet. Four wrappers Gravity Search Algorithm (GSA), Equilibrium Optimizer (EO), Gray Wolf Optimizer (GWO), and Binary Bat Algorithm (BBA) are used. The selected features are then classified using RF and SVMs algorithms. The DL-based approach achieves an impressive accuracy of 96.32% and a kappa value of 0.98, with 250 sort out features. The results were acquired from the Kaggle competition through the APTOS dataset, where the highest kappa value was 0.93. The DL-based approach

has a limitation of low performance when operated on huge datasets. In future work, different wrapper methods can be employed to extend the applicability of this approach to various types of diseases.

A novel feature map global channel attention technique (GCA) is introduced by Yang et al. [92] for early DR detection. Image preprocessing techniques such as Gaussian Filter, Adaptive Gamma Correction, CLAHE are applied to the FI. Within the GCA module, a deep CNN model called GCA-EfficientNet (GENet), is a single-dimensional convolution kernel built upon the topography of the feature map. The proposed model achieved final results of 95.6% for sensitivity, 98.9% for specificity, 95.6% for accuracy, and 95.6% for precision on the DR dataset from the Kaggle competition.

A segment-based learning strategy for DR identification, CNN-based segment-level DRE-classifier is proposed by Math et al. [93] by identifying the internal lesions using the mutually learning classifiers. To be more specific, the pre-trained convolutional network is adapted to produce segment-level DR estimation (DRE). The final classification of the DR image is obtained by combining the segment-level information from the DR Map (DRM). The model was trained using a publicly available dataset from Kaggle. The Kaggle dataset yields an AUROC curve for DR image recognition of 0.963, along with sensitivity and specificity of 96.37% and 96.37% on the higher sensitivity and specificity, respectively, outperforming substantially better.

A Inceptionv4 with Hyperparameter Optimization (I-v4-HO) is designed by Shankar et al [76] for the classification of DR in color FI. During the preprocessing stage, the contrast of FI is enhanced using the CLAHE model. After this enhancement, the preprocessed image undergoes segmentation through a histogram-driven segmentation approach. The HPTI-v4 system is then utilized for essential feature extraction from the segmented images, and these features are subjected to classification via a multilayer perceptron (MLP). The model's validation was conducted using the MESSIDOR dataset, showcasing improved performance with maximum values of 99.47% specificity, 97.91% sensitivity, and 99.12% accuracy. The proposed model suffers with the limitation of excessive time consumption. Future enhancements to the proposed model could involve the incorporation of advanced classification models.

E. DR SEGMENTATION

Segmentation is an essential computer vision task that extends from the idea of object detection. Image segmentation involves dividing an image into distinct sections at pixel level and labeling each section outlining the object boundaries within the image and assigning them specific classes accordingly. DR segmentation partitions the image into sections with respect to the features of DR such as EX, HM, and MA.

To enhance image segmentation, the system undergoes training using pre-segmented datasets, which may be created manually or obtained from open datasets. The research studies conducted for DR segmentation using ML and DL techniques are illustrated in Table 7.

1) ML FOR SEGMENTATION

A hybrid model is proposed by Udayaraju et al. [28] for OCT image classification and segmentation that combines the U-Net and MSVMs. Without the necessity for inner retinal layer segmentation, the method effectively identifies four DME classifications (mild, normal, moderate, and severe) using a hybrid technique based on GLCM and MSVM classifiers. The conditions affecting intra-retinal layers give rise to the complexity, which makes layer boundaries the layer limits, which inturn makes segmentation harder. According to the experimental findings, the proposed U-Net-MSVM offers a good classification accuracy of 98.37% in diagnosing retinal disorders on OCT images from MICCAI RETOUCH Group datasets. This method has faced the key challenge of reduced performance due to smaller datasets limiting robustness, and its high false positive detection rate.

2) DL FOR SEGMENTATION

Gao et al. [95] have developed a streamlined algorithm using BiSeNet V2 for segmenting multiple lesions in DR. They have incorporated a hybrid attention module into the semantic branch of BiSeNet V2 to enhance the extraction of local key features by adjusting weights on deep feature maps. Additionally, a ghost feature-mapping unit is utilized to optimize traditional convolution layers and decrease computational costs. The model employs a novel loss function based on dynamic thresholding to guide training, which adjusts weights for difficult samples with high loss and improves the model's focus on small targets. An ablation study on the IDRiD dataset confirms the effectiveness of each component, and the proposed model, BiSeNet V2-Pro, is evaluated against several state-of-the-art models. Compared to the baseline BiSeNet V2, BiSeNet V2-Pro demonstrates improvements of 11.44% in Intersection over Union (IoU), 8.49% in Dice coefficient (DICE), and 12.17% in sensitivity (SEN), with IoU for MA reaching 57.16%. Future research could explore self-supervised pre-training techniques to further enhance the model's performance and generalization across different domains.

Wang et al. [96] have introduced a fully automated deep CNN approach for the simultaneous segmentation of four different types of fundus lesions related to DR. A cooperative design featuring a contextual branch and a local branch was introduced to effectively handle multi-scale image data. To achieve a thorough integration of key characteristics from both branches, the authors also included an attention mechanism that successfully merged map features from all decoding layers. Moreover, a novel auxiliary classification task reduces overfitting and enhances lesion segmentation accuracy. The proposed method achieves mean AUC values of 67.7%, 62.9%, and 58.1%, respectively. via thorough

testing on three publicly accessible fundus datasets such as IDRiD, E-ophtha, and DDR. Small lesion identification is still a difficult task, and to ensure the ability to generalize the approach across additional groups, it has to be experimentally validated on large datasets. For future work, more emphasis must be given to collecting more FI to evaluate the segmentation algorithm.

Radha et al. [97] introduce a method called DPA-U-Net, which incorporates depth-wise parallel attention into the U-Net framework. This approach adds a depthwise convolutional layer to the downsampling path and a parallel attention system applied to the upsampling pathway of U-Net. By combining depthwise convolution with global information embedding (GIE), the model enhances its ability to capture intricate channel-specific details and addresses the typical problem of information loss found in conventional convolution and downsampling techniques. Additionally, the parallel attention network improves the retrieval of channel and spatial information from the decoder-encoder throughout the upsampling stage. Testing on three openly available datasets (STARE, DRIVE, and CHASE_DB1) confirms the effectiveness of the proposed model. The performance metric values for each dataset are as follows DRIVE specificity of 98.82%, Sensitivity of 81.44%, Accuracy of 98.82%, AUC of 98.97%, STARE specificity of 98.56%, sensitivity of 98.16%, accuracy of 98.65%, AUC of 99.02%, CHASE_DB1 specificity of 98.79%, sensitivity of 82.26%, accuracy of 98.03%, AUC of 99%. The existing work is responsive to correlations among channels in the input data which restrains the model's capacity to pick up on and accurately depict high-level characteristics that depend on interactions across various channels.

An automated approach for DR classification is introduced by Jagadesh et al. [98], to address the existing challenges through a two-pronged strategy. To handle the issue of imbalanced data with a small number of positive cases, they use an advanced version of the improved contoured convolutional transformer (IC2T) for segmenting blood vessels and optical discs (ODs). The authors have developed a specialized module for detecting contoured ODs and BVs, along with a dual convolutional transformer block that merges local and global contexts to establish robust associations. Following this, a second-stage Improved Coordination Attention Mechanism (ICAM) network is trained to identify retinal biomarkers for DR, such as MA, HM, and EX. The experimental results show average accuracies of 97%, 96%, and 98% on the MESSIDOR-2, EyePACS-1, and DIARETDB0 datasets, respectively, highlighting its advanced performance in the field. Future research could focus on enhancing and optimizing the IC2T model and the Rock Hyrax Swarm-Based Coordination Attention Mechanism.

Jebaseeli et al. [54] suggest that improving the segmentation of retinal blood vessels can enhance DR diagnosis. The author propose a DL-based approach designed to improve segmentation quality in diseased RI. The framework includes the Tandem Pulse Coupled Neural Network (TPCNN) model for automatic feature vector generation, a DL-Based SVM (DLBSVM) for blood vessel extraction and classification, and CLAHE for background removal and enhancement of blood vessel pixels. The suggested methods are evaluated using the DRIONS, DRIVE, HRF, STARE, and REVIEW datasets. The results indicate that their method achieves 99.54% specificity, 80.61% sensitivity, and 99.49% accuracy. Future research could investigate the inclusion of vessel tortuosity measures to provide a more detailed differentiation of the disease.

Sangeethaa et al. [61] have formulated a CNN architecture for automatically diagnosing the severity of DR. The method incorporates image pre-processing techniques such as enhancing contrast, applying thresholding, removing backgrounds, extracting the green channel, detecting contours, and performing morphological operations. Using high-performance GPUs, the network is trained on openly available datasets like DIARETDB0, DRIVE, and DIARETDB1_v1, as well as images from the Aravind Eye-Hospital, Coimbatore, India. The deployed CNN achieves 96.9% specificity, 93% accuracy, and 98% sensitivity. However, the model faces challenges in accurately classifying mild and moderate DR cases at early stages. Future work could address this by evaluating vascular curvature and additional textural features such as HM, MA, and EX, and by enhancing the CNN with a larger training dataset.

F. DR EARLY DETECTION

Early identification is crucial in managing serious conditions in DR effectively resulting in preventing severe vision impairment.Table 8 presents the various ML and DL approaches studied for the early detection of DR.

1) ML FOR DETECTION

Taifa et al. [34] present a novel DR detection method that integrates ML algorithms with deep feature extraction techniques. The proposed model is trained using the APTOS 2019 dataset, employing preprocessing steps such as the Gaussian filter for coherence, and image transformations like resizing, rotation, and color adjustments thereby boosting the training efficiency. The suggested method extracts features indicative of DR patterns in RI through three previously trained DL models: InceptionResNetV2, DenseNet121, and MobileNetV2. These models are combined with several ML classifiers, the reordered list: KNN, RF, SVM, MLP, LR, ET, SGD, ridge classifier, Passive Aggressive (PA), perceptron, Nearest Centroid (NC), and DT. The hybrid model achieved 95% accuracy in multi class classification and 98.36% accuracy in tasks involving binary classification. The model has several challenges such as limited sample sizes, dataset biases, and computational constraints.

The early detection of DR using ML models by grading the severity of diabetics is discussed by Avleen et al. [27].

TABLE 7. Summary of DR segmentation techniques.

Applications	Ref. Paper	Method	Dataset	Performance Metric	Limitation
Segmentation	[95]	BiSeNet V2-Pro	IDRiD	11.44% in Intersection over Union (IoU), 8.49% in Dice coefficient (DICE), and 12.17% in sensitivity (SEN), with IoU for MA reaching 57.16%	NA
	[28]	Hybrid technique based on GLCM and MSVM classifiers	OCT Images -MICCAI RETOUCH Group datasets	Accuracy 98.37%	Exclusively relying on this approach for identifying all retinal symptom conditions is suboptimal.
	[96]	A fully automated deep CNN	IDRiD, E-ophtha, DDR	AUC values of 67.7%, 62.9%, and 58.1%	Detecting small lesions remains challenging
	[97]	DPA-U-Net	DRIVE, STARE, and CHASE_DB1	Accuracy 98.82%	The model's challenge lies in accurately representing high-level features dependent on interactions between channels, influenced by sensitivity to channel correlations in input data.
	[98]	Improved contoured convolutional transformer (IC2T)	EyePACS-1, Messidor-2, and DIARETDB0	Accuracy 98%	NA
	[54]	Tandem Pulse Coupled Neural Network (TPCNN) model for automatic feature vector generation, a DL-Based SVM (DLBSVM) for blood vessel extraction and classification	STARE, DRIVE, HRF, REVIEW, and DRIONS	Accuracy 99.49%	NA
	[62]	CNN architecture	DRIVE, DIARETDB0, and DIARETDB1_v1	Accuracy 93%	NA

The study utilized publicly available and real-time datasets such as E-optha, MESSIDOR, and DiaretDb with 1361 colored FI taken from a colored 3CCD camera with a 45◦ field of view. The collected datasets were preprocessed for macula and OD detection using filtering and green channel extraction techniques. Subsequently, the textural features of the grayscale images were extracted using the GLCM. The experimental study was conducted using KNN, SVM, and DT for grading EX and MA. The outcomes indicate that the grading method for EX achieved 92.1% of accuracy with KNN and SVM. As an alternative for MA, the decision tree model achieved an outstanding accuracy of 99.9% when compared to other existing works. The study highlighted the key challenges such as high error rate and huge time consumption while larger datasets are used. In the future, further developments can build upon the grading approach to encompass a combination of MA and EX. Furthermore, additional features such as HM and cotton wool spots also need to be incorporated to enhance the comprehensive detection of DR with improved data availability.

Mahmoud et al. [84] developed a hybrid inductive machine learning algorithm (HIMLA) designed as an automated system for detecting DR. The HIMLA algorithm evaluates colored FIs to classify them as either unhealthy (presence of DR), or healthy (no DR), effectively identifying medical instances of DR. Four key stages are included in the algorithm: preprocessing, image segmentation, feature extraction, and classification. During preprocessing, normalization of the FIs is performed to enhance image quality. Image segmentation involves encoding and decoding to increase clarity. The multiple instance learning (MIL) approach is utilized for feature extraction and also for classification. The effectiveness of the HIMLA algorithm was assessed using the CHASE datasets for DR detection, with 95.31% sensitivity, 96.88% specificity, and 96.62% accuracy being achieved. It has been demonstrated that HIMLA outperforms other DR models, including DL Algorithm DR (DLADR) ML-based neovascularization Detection in the OD (MLBNVD), Genetic Algorithm-Based DR (GAB-DR), and Diagnostic Assessment-Based DL for DR (DADDR) thereby reducing the risk of vision loss. Further improvements are required for feature extraction and classification.

Abdel Salam et al. [99] has introduced a model to diagnose early NPDR using OCTA. The authors aim to examine the macular vascular network in both normal individuals and those in the early stages of NPDR using OCTA images and multifractal geometry analysis. The model has been trained using the SVM algorithm on 170 eye images collected at the Opthalmology Center at Mansoura University, Egypt. This model performed well with the highest accuracy of 98.7%. Image quality was the serious drawback in extracting the optimal features resulting in the desired result. Further enhancement can be done with a focus on the graphical user interface (GUI) making the model capable of extracting certain image attributes for early diagnosis, detection, and also identification of malignancy.

An integrated ML model is developed by Thulkar et al. [36] for Exudate (EX) Detection. The study integrates ML, data analysis strategies, and image processing to detect EX and assess DME severity, using datasets such as IDEDD Mumbai, IDRiD from Nanded, and DIARETDB1. A tree-based forward search algorithm is employed to identify the optimal subset of features, while the SVM and RF algorithms are utilized for feature classification. Image processing techniques such as feature scaling, masking, and resizing are used to segment the macula, optic disk, and vascular mesh in severity grading, which uses IDEDD data to achieve 100% specificity and 97.1% sensitivity. The developed model struggles with blurred images and cannot detect the retinal thickness, impacting its accuracy. Future enhancements include increasing teleophthalmology applications for more efficient early diagnosis and treatment, incorporating OCT images and employing DL for better feature extraction, and implementing multiclass classification in larger datasets.

2) DL FOR DETECTION

Jabbar et al. [37] proposed a DL approach to classify DR by varying severity levels. The authors utilized ResNet and GoogleNet models optimized using adaptive Particle-Swarm-Optimizer (APSO) for improved feature mining. The proposed model APSO-GRESNET is trained on the Kaggle EyePACS1 dataset. Image processing techniques such as image resizing, green channel extraction, Tophat Transformation (White Tophat), and BottomHAT Transformations (Black Tophat) were used. The APSO-GRESNET hybrid model performs well by combining the strengths of Google's ResNet with APSO. This method outperformed others with an accuracy of 94% on the benchmark data-set preprocessing approaches to remove noise and artifacts successfully.

Prabhakar et al. [71] introduced a DL approach based on optimization techniques using FI for detecting DR. Initially, the FI undergoes preprocessing steps involving RoI and median filtering extraction to eliminate noise. Subsequently, lesion segmentation is implemented using U-Net, trained with a novel optimization algorithm called Gannet Pelican Optimization Algorithm (GPOA), which integrates the Pelican Optimization Algorithm (POA) and Gannet Optimization Algorithm (GOA) to identify various lesion types. The IDRiD is used for model training, which is later subjected to data augmentation methods such as Rotation, shearing, cropping, translation, and flipping are applied to augment the dataset, followed by feature extraction from both image and vector-based FI features. Additionally, Deep Q-Network (DQN) is utilized for DR recognition, trained with the Exponential Gannet Pelican Optimization Algorithm (EGFOA), which combines Exponentially Weighted Moving Average (EWMA), GOA, and Firefly Optimization Algorithm (FFA).

The outcome indicates a maximum accuracy of 91.6%, sensitivity of 92.2%, and specificity of 91.9%.

Shamrat et al. [100] conducted a study to automate the classification process of DR at different stages using CNN models. The researchers evaluated the performance of fifteen pre-trained models alongside the newly proposed DRNet13 model to determine the most effective model for accurately staging DR based on FI that encompass five DR classes. They employed image preprocessing techniques, such as median filtering for noise reduction and Gamma correction for image enhancement. To improve the generalization capability of the training model, they expanded the DR dataset from 3662 to 7500 images by applying Gaussian filtering to the 224×224 filtered retina scans obtained from the Aravind Eye Hospital in India. The researchers used feature maps to identify decision-making areas within the DRNet13 model. The evaluation of the model showed a 97% accuracy for DR detection, surpassing other CNN architectures in terms of speed and efficiency. One limitation of this study is the representativeness of the dataset. To enhance the robustness of the DRNet13 model, future efforts could focus on incorporating diverse data that covers a wide range of DR phases and patient demographics.

Thomas et al. [32] have proposed an approach which focuses on utilizing retina FI obtained from both publicly available datasets such as EyePACs and MESSIDOR, as well as clinical dataset from Chaithanya Eye-Hospital, Kerala. The initial phase involves noise removal and contrast enhancement of the input images. A Trilateral Filter is employed for noise lessening, followed by contrast enhancement using CLAHE. Next, Extended Piecewise Fuzzy C-Means Clustering (EPFCMC) Method isolates the thick blood vessels from the improved image. GLCM features are extracted from the segmented image, followed by feature selection using the War strategy optimization. Finally, an ensemble transfer learning (TL) approach is used, employing CNN and SVM classifier models to classify the given FI as either DR or normal. This method achieves an impressive accuracy of 98.94% with the CNN and SVM classifier model. The proposed approach is limited by its focus on precisely identifying thick blood vessel regions and struggles with irregularly shaped vessels and boundary pixels in FI. Future research directions may involve exploring the usage of the EPFCMC method and adopting a super pixel-based clustering approach.

Shoaib et al. [101] proposed a DR diagnosis method utilizing advanced DL techniques. The approach is based on transfer learning (TL), using pre-trained models like Inceptionv3 and InceptionResNetv2 to derive features and adjust specific layers for the diagnostic task. DiaCNN, specifically designed for eye disease classification, was introduced by the authors. The Ocular Disease Intelligent Recognition (ODIR) dataset, comprising eight different eye disease categories, was used to assess the method. The outcomes were promising: leveraging transfer learning, the InceptionResNetv2 model achieved an impressive 97.5% accuracy in both the training and testing phases. The Inceptionv3 model reached even higher accuracy, achieving 99.7% during training and maintaining 97.5% in testing. The DiaCNN model demonstrated outstanding performance, with 100% accuracy in training and 98.3% in testing. Future work will focus on developing noise-resistant systems, exploring feature fusion techniques for improved classification, and investigating DL approaches combined with segmentation strategies.

Guefrach et al. [102] create a graphical user interface that integrates DL model for automated DR diagnosis. Currently, identifying DR from computerized FI presents a challenge in medical image processing, necessitating the development of new strategies. Manual analysis of RFI is time-consuming and requires considerable expertise. To assist healthcare professionals, this study creates a user-friendly interface that integrates imaging algorithms to evaluate DR indicators in patients' FI. The research utilizes the APTOS 2019 dataset, employing a deep neural-network for diagnosis, particularly the ResNet152-V2 model. This model has achieved perfect performance, with 100% accuracy across all evaluation metrics, including recall, F1 Score, precision and accuracy. The performance of this model may be limited by mobile-acquired images.

A DR screening method using asymmetric DL Features is introduced by Jena et al. [33]. This procedure entails extracting these features by employing a U-Net for segmenting the blood vessels and optical disc. The effectiveness of this method was assessed using two publicly available RI datasets, namely MESSIDOR and APTOS. The authors choose the green channel in DR screening, addressing non-uniform brightness and low contrast in color FI by enhancing its contents through CLAHE. Following this, the enhanced green image underwent preprocessing. Subsequently, a combination of CNN and SVM is employed for the classification of DR lesions, which are categorized into four distinct classes: normal, HM, MA, and EX. The accuracy rates for identifying non-DR in the datasets are reported as 98.6% and 91.9%, respectively. In the case of detecting EX, the accuracy is noted as 98.3% and 96.9%, HM an accuracy of 95.5% and 93.6%, MA is detected with 98.0% and 95.4% accuracy for the MESSIDOR, and APTOS dataset. Non-diabetic cases (normal eyes) are classified with high accuracy, achieving a sensitivity of 99.35% and 93.65%. The study proposes further exploration into U-Net and DL-based classification methods for other medical image segmentation tasks to improve clinical support.

An automated DL-based model HemNET for HM detection is proposed by Aziz et al. [82], which may be useful to ophthalmologists as a further instrument to help them accelerate and simplify conventional screening procedures. Improving the quality of images and locating potential bleeding sites are part of the preprocessing step. Gradient information is used to dynamically modify

TABLE 8. Summary of early detection techniques.

Application	Ref. Paper	Method	Dataset	Performance Metrics	Limitation
Early Detection	[34]	DL models: Inception-ResNetV2, DenseNet121, and MobileNetV2, Ml models: KNN, RF, SVM, MLP, LR, ET, SGD, ridge classifier, Passive Aggressive (PA), perceptron, Nearest Centroid (NC), and DT	APTOS 2019	Accuracy 95%	NA
	[37]	APSO-GRESNET	Kaggle EyePACS1	Accuracy 94%	
	[71]	GPOA based U-Net	IDRiD database	Accuracy 91.6%	NA
	[100]	DRNet13 model	Diabetic retinopathy dataset	Accuracy 97%	NA
	[32]	An ensemble transfer learning (TL) approach is used, employing CNN and SVM classifier models	Messidor, EyePACs dataset and the in-house clinical dataset from Chaithanya-Eye Hospital Kerala	Accuracy 98.94%	Irregularly shaped vessels, thin vessels, and edge pixels
	[101]	DiaCNN	Ocular Disease Intelligent Recognition (ODIR) dataset	Accuracy 100%	NA
	[102]	ResNet152-V2 model	APTOS 2019 Blindness Detection Database	Accuracy 100%	NA

TABLE 8. *(Continued.)* **Summary of early detection techniques.**

[27]	KNN, SVM, and DT	Messidor, DiaretDb and Eoptha	Accuracy 99.9%	It cannot grade both EX and MA, and excludes cotton wool spots and HM for diabetic retinopathy detection.
[84]	Hybrid inductive machine learning algorithm (HIMLA)	CHASE datasets	Accuracy 96.62%	NA
[33]	Hybrid CNN and SVM	APTOS and MESSIDOR	The accuracy rates for identifying non-DR in the datasets are reported as 98.6% and 91.9%, respectively. In the case of detecting EX, the accuracy is noted as 98.3% and 96.9%, HM an accuracy of 95.5% and 93.6%, MA is detected with 98.0% and 95.4% accuracy for the MESSIDOR, and APTOS dataset	NA
[82]	HemNET	DIARETDB121 and DIARETDB022	Sensitivity 94.21%, Specificity 94.76%	NA
[103]	DLS	SM1 and SM2 datasets	AUROC of 0.99, specificity 96.00% and the sensitivity 93.86%	Focusing solely on one ethnic group may limit the algorithm's generalizability across diverse populations.
[67]	PCNN architecture	KAGGLE 2015 & APTOS 2019	Accuracy 91.78% and 97.27% for the respective datasets	NA
[77]	ICOA-DLDRD method	MESSIDOR	Accuracy 99.65%	NA
[104]	CNN-based model ResNet50, InceptionV3, DenseNet169, and DenseNet121	Asia Pacific Tele-Ophthalmology Society (APTOS-2019) dataset (Aravind Eye Hospital 2022).	Achieved accuracy values of 95.71%, 94.73%, 95.95%, and 96.64% respectively	NA

TABLE 8. *(Continued.)* **Summary of early detection techniques.**

	[63]	CNN-ADL	EyePacs	Accuracy 99.66%	Identifying the five severity levels of diabetic retinopathy is challenging due to complex lesion structures, especially in color fundus images.
	[38]	RF model	Clinical dataset - DR risk was examined in 172 people with type II diabetes, SRM Hospital	Accuracy 94.9%	The study focused solely on Type II diabetic patients with a small sample size and without consideration of other coexisting diabetic conditions.
	[83]	MdFCM, CNN	DIARETDB1, APTOS, Liverpool dataset	Accuracy 98.6%	NA
	[99]	SVM	170 eye images have Opthalmology Center at Mansoura University in Egypt.	Accuracy 98.7%	Challenges in image quality hinder accurate feature extraction; OCTA images below 30% quality pose diagnostic difficulties for physicians.
	[105]	VGG-NiN model	EyePACS	Accuracy 80%	NA
	[36]	SVM and RF	IDEDD-Metropolitan region of Mumbai, IDRiD-Nanded & DIARETDB1	Specificity 100% and Sensitivity 97.1%	NA
	[106]	TL-based CNN architecture	Kaggle dataset: 3050 images, 4 classes: A(1805), B(370), C(999), D(295).		NA
	[107]	Hybrid PCA–firefly deep neural network model	Diabetic Retinopathy Debrecen dataset from UCI machinelearning repository	Accuracy 97%	Performance may degrade with low-dimensional datasets, potentially leading to overfitting of the model.
	[108]	CNN	MESSIDOR and IDRiD	Accuracy 90.89%	NA

the lighting of FI using a modified gamma correction approach, which efficiently addresses uneven brightness levels. The technique uses a mix of mathematical morphology, Gaussian match filtering, and entropy thresholding to identify HM possibilities. Metrics including accuracy, precision, sensitivity, and specificity are used to evaluate the models performance on two datasets DIARETDB1 and DIARETDB0. The suggested model works better than VGG-16, LeNet-5, AlexNet, ResNet50, and even the least complicated network. The HM network's workflow is further assessed in terms of classification accuracy and training time, showing promising classification accuracy while lowering training time considerably.

Nunez et al. [103] A DL system (DLS) has been created by Nunez et al. [58] to detect DR that can be referred from RI captured by the means of handheld-non-mydriatic fundus cameras. A risk factor-based LR model and the DLS were trained and cross-validated using a SMARTIndia1, SMART-India2 dataset from 16,247 eyes (including 9,778 individuals) that included macula-centered and optic disc-centered images. Using two-field RI, the DLS showed remarkable performance with an AUROC of 0.99 (with a 95% confidence range of 0.98–0.99). At the operational point of Youden's index, the specificity was 96.00% (ranging from 94.68 to 98.09%) and the sensitivity was 93.86% (varying from 91.34 to 96.08%). The DLS attained an AUROC of 0.96 (range from 0.95–0.98) for the optic-disc field and 0.98 (ranging from 0.98–0.98) for the macula field when given single-field inputs. A sensitivity of 90.01% (range from 88.95–91.01%) and a specificity of 96.09% (ranging from 95.72–96.42%) indicate the highest complement performance. The study's restriction to a mono-ethnic population may limit the algorithm's generalizability and variability in retinal image curation for diverse populations in DR screening.

Nahiduzzaman et al. [66] developed an automated technique for DR detection using CLAHE-preprocessed FI, to enchance lesion visibility. For feature extraction, a parallel CNN (PCNN) was utilized, and for classification, an extreme learning machine (ELM) technique for DR was applied. The PCNN architecture decreases the amount of time needed for feature extraction by having fewer layers and parameters than a comparable CNN structure. The 34,984 FI from the Kaggle DR 2015 competition dataset and the 3,662 FI from the APTOS 2019 dataset were the two datasets utilized to assess the efficacy of the technique. In particular, the suggested approach produced an accuracy of 91.78% and 97.27% for the respective datasets. In evaluating the classification performance, they took into consideration AUC, accuracy, F1-Score, and recall. Particularly, the method outperformed previous approaches in terms of prediction time, model complexity, and classifier performance.

Parthiban et al. [77] presented a DL-based method for RFI interpretation that combines the Coyote Optimization Algorithm with DL-based DR Detection and Grading (ICOADLDRD). The principal objective of the ICOADLDRD approach is to detect the presence of DR in these RFI pictures. The program uses an optimal region-expanding segmentation approach for segmentation and Gabor filtering (GF) for noise reduction. The initial seed locations and segmentation thresholds are obtained by applying the Glowworm Swarm Optimization (GSO) method. Additionally, a SqueezeNet model with a Class Attention Learning (CAL) layer is used to extract the feature vectors. Lastly, the Coyote Optimization Algorithm (COA) and a Deep Extreme Learning Machine (DELM) classifier are used for DR recognition and grading. To be more precise, gamma and the penalty parameter C in the DELM model are adjusted using COA. With the greatest accuracy of 99.65%, the ICOADLDRD approach is assessed using the benchmark MESSIDOR dataset.

A DL-based technique that uses FI to identify the various phases of DR has been presented by Naik et al. [104]. The proposed model has undergone training and was able to categorize DR into classes. To categorize DR from FI, a CNN-based model with data augmentation is therefore proposed. The APTOS-2019 dataset from Aravind Eye Hospital is graded into five classes (0 to 4), representing different stages of DR is used for the model training. Several top-tiered GPU trained models, namely, ResNet50, InceptionV3, DenseNet169, and DenseNet121 achieved accuracy values of 95.71%, 94.73%, 95.95%, and 96.64% respectively.

An approach called active DL (ADL) is introduced by Ozbay et al. [62], utilizing a CNN-ADL multilayer design for the automatic identification of different stages of DR. The artificial bee colony (ABC) technique is used for image segmentation during preprocessing in order to enhance lesion identification in the ADL system. The threshold value for this approach is derived from the picture histogram. Additionally, the ADL-CNN architecture, an effective CNN architecture, is created to automatically extract segmented retinal characteristics. The study utilized the EyePACS dataset, funded by the California Health Care Foundation, which comprises 35,122 RIs with various severity of DR, to train the CNN-ADL model. The CNN-ADL model outdoes other methods such as VGG16-FC1, VGG16-FC2, VGG16, and Inceptionv3 on the same dataset, achieving significantly higher accuracy results in identifying lesions and categorizing severity levels of DR. The CNN-ADL model effectiveness is evaluated through performance metrics that achieve outstanding performance with: 94.58% F-measure, 99.66% accuracy, 93.76% sensitivity, and 96.71% specificity.

Zubair et al. [105] have proposed an approach that focuses on classifying different stages of DR using a DL model. VGG16 is integrated with Spatial Pyramid Pooling (SPP) and Network-in-Network (NiN) to prioritize the categorization of DR stages. After optimizing for fewer learnable parameters, the final VGG-NiN model achieves better nonlinearity and scale invariance. A validation of the model's enhanced classification ability across various scales of DR images is carried out through evaluation of the EyePACS and Kaggle datasets, employing measures including accuracy,

specificity, F1-Score, and recall. The use of an ensemble strategy based on DL advances the work on the identification of DR. Results on unbalanced Kaggle datasets provide better performance than the state-of-the-art non-ensemble and ensemble approaches, demonstrating the computational efficacy and efficiency of the suggested methodology. The proposed ensemble VGG-NiN model has shown an accuracy of 80%. Future work can be done on the model architecture and preprocessing techniques

Naz et al. [83] developed a technique for early recognition of DR using a combined approach of unsupervised and DL. The preprocessing steps include normalization of the data and later, features such as ridge density, ridge breadth, and valley breadth, are computed. This study focuses on the prevalence of DR and its impact on blindness worldwide in order to address the urgent need for automated DR identification. Initially, a modified fuzzy clustering method (MdFCM) is built with the revision of the objective function of fuzzy clustering method and k-means. The features are then consolidated with the help of a proposed modified CNN for better outcomes. comparison of results across three different datasets including APTOS, DIARETDB1, and Liverpool, were utilized alongside fuzzy clustering methods, and k-means for evaluating the model's applicability across different datasets is done achieving a remarkable 98.6% accuracy rate, the proposed approach marks a significant advancement in early-stage DR detection.

A structured multi-task DL system has been proposed by Juan et al. [109] to simultaneously detect characteristics associated with DR and diagnose the severity of DR in FI. To take into consideration the link that exists between DR severity levels and DR-related traits, the framework has a hierarchical structure. Dataset used consists of a total of 89917 digital FI in the experiments, all of which were collected by Shenzhen SiBright Co. Ltd. (Shenzhen, Guangdong, China). A variety of assessment measures, such as precision-recall analysis, receiver operating characteristic analysis, and quadratic weighted Cohen's kappa coefficient, were used to evaluate the proposed technique on two different testing sets. Furthermore, to evaluate the effectiveness of the suggested method against general ophthalmologists with different degrees of expertise, a grader research was carried out. In assessing performance, they have taken into account AUC and AP. DR Severity Level Diagnosis achieved higher kappa values of 0.8083 (std = 0.0097) for validation set, 0.7421 (std = 0.0172) for Test1, and 0.7187 (std = 0.0183) for Test2. Referable DR Diagnosis achieves best AUC values for the validation set (0.9537), Test1 (0.9646), and Test2 (0.9456). Although the existing hierarchical multi-task DL architecture works exceptionally well in simultaneous deep recovery diagnostics, further attempts will address the remaining challenges. With a focus on ongoing performance enhancement, future work can be done to strengthen the model's robustness across a variety of datasets and increase its suitability for clinical usage in the real world.

Samanta et al. [106] proposed a robust TL-based CNN architecture for analyzing color FIs. The FI is preprocessed using image preprocessing techniques such as contrast enhancement, thresholding, CLAHE, and weighted Gaussian blur. The proposed model is built by stacking convolution layers in which the major distinctions and curves are classified in the first layers while the HE and HM are classified in the last layers. DenseNet121 model was chosen for its efficiency from the existing CNN architectures for classification. Finally, the fully connected layers were removed, and the output was flattened to one dimension. The proposed model excels on 3050 training images and 419 validation images obtained from the publicly available Kaggle dataset, effectively Classifying diabetic retinopathy from texture, blood vessels, and HE. The model classifies FIs into 4 groups: PDR, Mild DR, Moderate DR and No DR. The publicly available dataset with 3050 images across 4 classes (A: 1805, B: 370, C: 999, D: 295) was trained on Google Colab, achieving Cohen's Kappa scores of 88.36% on the validation set and 98.09% on the training set. The model's possible overfitting due to the skewed class distribution and potential lack of generalizability to larger and more different datasets remain a limitation. Future work will apply semantic segmentation to classify retinal arteries and vessels to comprehensively understand retinopathy patterns.

Thippa et al. [107] implemented a PCA-Firefly-based DL Model for early detection of DR. The dataset utilized in this study is sourced from the publicly accessible UCI ML Repository. Initially, the standard scaler approach used to normalize the raw dataset, followed by utilizing PCA to mine the most vital characteristics of the dataset. The firefly techniques are utilized to further decrease its dimensionality. After that, a deep neural-network model is trained using this compressed dataset for classification purposes. This model has shown a good performance than expected in terms of measures like precision, sensitivity, specificity, accuracy, and recall. After thorough preprocessing, the hybrid PCA–firefly deep neural network model performed exceptionally well in the classification of DR of accuracy 97%. However, it could run into difficulties while working with low-dimensional datasets, which might indicate an overfitting issue. Subsequent investigations may examine its applicability in several high-dimensional fields.

A DL model is developed by Saranya et al. [108] for automatic recognition of NPDR. This model is aimed at the automation of the detection and grading of NPDR from RF images using CNN. MESSIDOR and IDRiD are the datasets used for training the model. Before applying the CNN, the images were preprocessed and the resolution was adjusted to (256 × 256). The proposed model exhibits an accuracy of 90.89% for the MESSIDOR dataset and 90.29% for the IDRiD dataset in the case of a binary classifier model and a sensitivity of 96% for the MESSIDOR dataset and 84% for the IDRiD dataset in case of a 4-ary model. Future research can advance by implementing diverse

preprocessing techniques prior to passing data through various computational layers.

G. DR EARLY DIAGNOSIS

DR early diagnosis through proactive management and regular screening is essential for preserving the vision of individuals with diabetes. Diagnosis is correlated with different biological factors affecting a patient's eye and thus AI can be effectively applied to diagnose DR at the very early stages considering various biological factors associated with DR. Table 9 illustrates the research studies done on DR Early Diagnosis with DL.

1) DL FOR DIAGNOSIS

Mohammadi et al. [78] explores the assessment of the functionalities of Vertex AI and ChatGPT to allow code free image pre-processing, data analysis, and ML model training for DR diagnosis. MESSIDOR-2 dataset having 1740 images is utilized for both ML models. The image processing for fundus photography images was done using the CLAHE technique, which was implemented using Fiji software. ChatGPT and Vertex AI successfully trained two ML models for DR diagnosis with high precision and recall utilizing a code-free approach. The initial model functioned as a binary classifier to diagnose DR, while the other model evaluated the severity of the condition. The first model, performed with an area under the precision-recall curve (AUPRC) of 81%, recall of 72.83% and with a precision rate of 81.81%. The second model gave a AUPRC of 0.90 with a precision and recall of 84.48%. The model has certain limitations such as ChatGPT might malfunction on encountering identical queries, and the same answer may not be provided always.

An optimized DL model is proposed by Venubabu et al. [79] for the diagnosis of DR. The main components of the proposed DR system involve preprocessing, extracting features, classifying, and segmenting. The RFI are RGB images applied in the preprocessing stage with the G-channel. CLAHE and HE were used to enhance the image quality. The OD was removed by the Circle Hough Transform (CHT) technique after which, blood vessels are removed using Gray Level thresholding. After that, the Modified Expectation Maximization (MEM) techniques were used to segment EX. The GLCM was used to extract the optimal features (GLCM). Finally, a Deep Neural Network with a Butterfly Optimization Algorithm (DNNBOA) classifier was used to classify the DR into phases. Using the MESSIDOR dataset, the model achieved an accuracy of 98.9% and the DIARETDB1 dataset recorded 98.3%. Particularly, the suggested model performs more accurately on the MESSIDOR dataset than techniques such as DNN, SVM, DBN, and KNN with differences in accuracy of 14.5%, 23.29%, 16.6%, and 25.9%. The presented model shortens computation times and improves system correctness. Future enhancements in speed can also be achieved by using an effective metaheuristic algorithm.

A bi-directional extended short-term memory model is proposed by Phridviraj et al. [80] which supports the integration of DL into the FI-based diagnosis procedure for DR detection. The author uses Multiscale Retinex with Chromaticity Preservation (MSRCP) for improving image contrast, and have utilized MESSIDOR 1200 images, IDRiD 413 images, and Kaggle 35126 images with binary classification requiring 10249 DR and 938 usual images, and multi-class classification employs various severity levels of DR images from these datasets. The model used for classification Bi-LSTM-MSRCP has shown an accuracy rate of 96.77%, and performs better. To increase interpretability, the study recommends additional study into FI and data augmentation inside Explainable AI frameworks. despite the limited resources, it is recommended to look at how data augmentation could improve RFI-based DR diagnostic models.

DR Prediction using DL is done by Alwakid et al. where they have provided methods for enhancement of the model performance using CLAHE and ESRGAN [67]. According to the authors, the suggested strategy provides two unique instances. In the first instance, images are improved by combining an enhanced super-resolution generative adversarial network (ESRGAN) with the CLAHE filtering technique. The second instance, on the other hand, excludes image augmentation. With the same settings, augmentation techniques were used to generate a dataset that was well-balanced for both cases. Using the APTOS datasets, Inception-V3 produced a model with an accuracy of 98.7% for the first instance and 80.87% for the second instance. Testing the suggested approach on large diverse datasets, examining the use of VGG or ResNet, DenseNet including additional augmentation techniques and separately analyzing the influence of CLAHE and ESRGAN on the classification process was the major focus of the work.

A DL model named Bat Algorithm (BA)-DeepForest identification of DR using FI was implemented by Modi et al. [81]. This method was structured around four stages: preprocessing, extracting features, selecting features, and classifying. In the preprocessing stage, noise removal and contrast enhancement are carried out. Noise elimination is achieved using noise channel and median filters, while the CLAHE algorithm is employed to enhance image contrast. Lesion regions are identified through a k-means-based segmentation technique. In the final step, appropriate features are utilized for the diagnosis and classification of DR by implementing the Deep-Forest cascade method, which provides a binary classification of DR. The BA-DeepForest model's effectiveness is assessed using two DR datasets. One dataset comprises 3200 fundus eye RI, and the other dataset is MESSIDOR. The outcomes are contrasted with various models such as VGG16, SVM, ANN, KNN, VGG19, Deep Forest and InceptionV3. The outcomes demonstrate that the suggested BA-Deep-Forest model achieves higher accuracy rates 92.94% and 94.61%, F1score rates 95.31% and 96.49%, specificity rates 94.95% and 98.04% and

TABLE 9. Summary of DR early diagnosis techniques.

Application	Ref. Paper	Method	Dataset	Performance Metrics	Limitation
Early Diagnosis	[78]	DL	Messidor-2	The first model gave a AUPRC of 81%, recall of 72.83% and with a precision rate of 81.81%. The second model gave a AUPRC of 0.90 with a precision and recall of 84.48%	NA
	[79]	DNNBOA	MESSIDOR & DIARETDB1	Accuracy of 98.9% and 98.3% respectively	NA
	[80]	Bi-LSTM-MSRCP	MESSIDOR	Accuracy 96.77%	NA
	[68]	Inception-V3	APTOS	Accuracy 98.7%	NA
	[81]	BA-DeepForest model	MESSIDOR	Accuracy 94.61%	NA
	[110]	Hybrid Fully Dense (FD)-CNN architecture with CNN, ML techniques (D-KNN, D-SVM)	The Duke OCT and University of California San Diego (UCSD) datasets	Average Accuracy 98.55%	NA
	[111]	XGBoost-EBM Hybrid Model	317 samples, three classes, 39 samples with imputed missing values	Accuracy 91.25%	External validity in a separate cohort involves DR classification using demographic, clinical, and metabolomic data from patients.
	[64]	ADL-CNN	EyePACS and MESSIDOR	Accuracy 73%	Accurately detecting SE, HE, MA, and HM poses challenges, particularly distinguishing Mild from Normal grades during training, and the model's performance is parameter-dependent.
	[112]	Inception-v3 model	Kaggle DR database	Precision 91.44%, Specificity 91.35%, and Sensitivity 92.51%	NA
	[109]	DL	89917 digital funds images in the experiments, all of which were collected by Shenzhen SiBright Co. Ltd. (Shenzhen, Guangdong, China)	kappa values of 0.8083 (std = 0.0097) for validation set, 0.7421 (std = 0.0172) for Test1, and 0.7187 (std = 0.0183) for Test2 and AUC values for validation set (0.9537), Test1 (0.9646), and Test2 (0.9456)	In the training set, about 2% of images show mild NPDR, and features like NVE, NVD, PFP, VPH, and TRD are rare in the testing set of this study.
	[72]	PMNPDR system	Idrid	Accuracy 97.8%	NA

sensitivity rates 95.67% and 97.39%, using the 10 fold cross-validation method. Identifying the five severity levels of DR in color FI may be difficult because of the complex lesion structures. Further creating an ADL-CNN and experimenting with different CNN parameters for DR and multimedia applications.

Abbood et al. [63] proposed a hybrid DL model which improves the value of color FI in DR detection. The author aims to address complex lesion structures through an ADL-CNN approach, utilizing ResNet50-based DL networks and image preprocessing techniques like shape cropping, cropping, and Gaussian blurring. The study evaluates the model on MESSIDOR and EyePACS datasets. Initial results show 73% accuracy, 77% sensitivity, and 56% Positive Predictive Value (PPV) for DR classification, which improves to 92% accuracy, 82% sensitivity, and 78% PPV

after model enhancement. Enhanced images yield further improvements to 93.6% accuracy, 84% sensitivity, and 76% PPV. The limitations of this work are detecting specific abnormalities and the similarity between Mild and Normal grades, alongside parameter dependency of the enhancement model. Future work may focus on addressing illumination correction and extension of model capabilities to improve diagnostic accuracy across all DR severity levels.

Ghouali et al. [112] introduce an AI-based smart teleophthalmology application aimed at diagnosing DR. The application integrates DL using TensorFlow's Inceptionv3 model to analyze FI sourced from the Kaggle DR database, advancing image recognition capabilities in medical diagnostics. The major aim was to improve the accuracy and efficiency of DR diagnosis, promoting mHealth and prompt treatment by clinicians. The Kaggle database comprises 88,702 color FI evaluated on a scale of 0 to 4 for DR severity according to Early Treatment Diabetic Retinopathy Study (ETDRS) standards. This AI application has shown a precision of 91.44%, specificity of 91.35%, and sensitivity of 92.51%. The generated reports have the potential to aid doctors in assessing the severity level of the disease in individuals. Future research could focus on utilizing AI-based devices for diagnosing myopia and Apea syndrome.

Juan et al. [94] introduced a complex DL system to detect DR diagnosis severity and characteristics concurrently. This method considers how DR severity relates to specific traits using a dataset of 89,917 digital FI from Shenzhen SiBright Co. Ltd. (Shenzhen, Guangdong, China). The author evaluated the system performance with quadratic weighted Cohen's kappa coefficient, precision-recall analysis, and receiver employ characteristic analysis on two different test sets and also compared it to general ophthalmologists assessments. The approach achieved significant results for the severity level of DR diagnosis and showed kappa values of approximately 81%, 74%, and 72% for the Test1, Test2, and validation set respectively. DR diagnosis achieved strong AUC values of about 96%,95% for Test1, Test2, and 95% for the validation set, respectively. With a focus on ongoing performance enhancement, future work can be done to strengthen the model's robustness across a variety of datasets and increase its suitability for clinical usage in the real world.

A model named Prognosis of MA and early diagnostics system for non-proliferative DR (PMNPDR) designed for identifying DR by forecasting MA and establishing an early diagnosis system for NPDR is presented by Qiao et al. [72]. The experimental assessments were conducted on the IDRiD dataset, with preprocessing executed through the optimal bandpass filter method. The author introduces the PMNPDR system, which develops a DCNN for semantic separation of FI to upgrade the accuracy and efficiency of NPDR detection, with a focal emphasis on DL, and harnesses GPU acceleration for the identification and segmentation of MA in FI. This automated system assists ophthalmologists in grading FI, distinguishing between severe NPDR, moderate NPDR and early NPDR. The model's evaluation has demonstrated an accuracy of 97.8% where an extensive training set not only entails a considerable time investment but also introduces challenges related to class imbalance.

H. DR GRADING

AI can help healthcare professionals to perfectly grade the stages of DR on the basis of a patient's retinal medical condition taking into account the features of DR from the RI. The research articles relevant to DR Grading in DL are shown in Table 10.

1) DL FOR GRADING

A significant vision ailment that many diabetes individuals experience is DR, requiring effective diagnostic methods. To extract features from FI and identify DR, prior research has employed DL approaches such as sparse coding, auto-encoders, and limited Boltzmann machines.

DR, a microvascular condition linked to increased cardiovascular disease risk, presents a significant global healthcare challenge. Baba et al. [113] introduce an innovative approach for detecting DR the DR-Residual Network (DR-ResNet+). Using DL, this model autonomously extracts features and achieves optimal results within seven training epochs. To validate its robustness, simulated outcomes are compared with established DL approach like VGG16, GoogleNet, and AlexNet. This comparison makes use of a kaggle dataset that includes over 35,000 RIs, providing a comprehensive dataset for analysis. Furthermore, the proposed model is validated using external datasets such as IDRiD and MESSIDOR. The data is preprocessed using the image processing techniques of resizing and normalization. Hyperparameter optimization utilizes grid and random search methods to achieve the best possible outcome. Simulation outcomes demonstrate outstanding performance with an Sure, rearranged as requested, here is the comma-separated list: 0.9898 accuracy, 0.9916 specificity, 0.9670 precision, 0.9829 sensitivity, 0.9748 F1-score. The further research extension may involve integrating various medical features, such as medical history, patient demographics, laboratory findings, or genetic markers.

However, a recent study by Batool et al. [114] offered a unique strategy employing EfficientNet batch normalization (BNs) pre-trained models, which are a form of CNN, to automatically generate distinguishing characteristics from FI. The model is trained on the datasets DeepDRiD and Kaggle EyePACS. For preprocessing, the transformations used are foreground cropping, resizing, and gaussian smooth. Compared to previous methods, this study outperforms DeepDRiD by achieving F1-scores above 80% for all EfficientNet BNs on the EyePACS dataset. The study recognizes limited exploration in achieving a higher F1 score for distinguishing all DR groups, highlighting challenges in achieving comprehensive accuracy across different severity levels of the condition. Enhancing DR grading performance may involve integrating advanced preprocessing, data

augmentation, high-resolution image utilization, ensemble techniques, and additional features into efficient net BN models.

The influence of DL algorithms for DR on physicians in computer-assisted environments is comprehended by Sayres et al. [115]. The authors have used the Inceptionv4 model for classification. Training images were sourced from EyePACS clinics and three eye hospitals in India, specifically Aravind Eye Hospital, Sankara Nethralaya, and Narayana Nethralaya, where patients were undergoing DR screening. For grading, 1612 diabetic patients from 1796 RFI were collected. A group of eye specialists evaluated images for DR rigorousness under circumstances: grades plus heatmap, unassisted, or with grades only. Specifically for the Inceptionv4 model for classification, the mean sensitivity improved from 79.4% (unassisted) to 87.5% (grades only) and further to 88.7% (grades plus heatmap), while specificity remained high across all conditions. The limitation of this work is methodology tends to be either overly aggressive or excessively cautious in emphasizing image regions. As a future work, the degree to which the selection of operating points can impact the balance between misses and false referrals among clinicians can be examined, and the considerations regarding gradability can be evaluated.

I. SUMMARY

A study on the Application of AI in DR is done considering classification, segmentation, Early Detection, Early Diagnosis, and Grading. AI is employed in various capacities in the diagnosis of DR. Every specific step in the detection of DR is focused in the light of AI technologies for an efficient diagnosis of the disease. Several works are found in the classification of DR, where the dataset is thoroughly processed by AI technologies for better accuracy. Some works are done specifically on segmentation. The art of segmentation is vital in the detection of abnormalities in an RFI. Efficient segmentation techniques are proposed based on the existing state-of-the-art techniques overcoming several drawbacks. Some other works are concentrated on the areas of Early Detection, Diagnosis, and Grading especially to highlight the thin line of difference between different levels of DR. AI can be further applied even at a larger extent in the diagnosis of DR with the identification of trivial yet prominent aspects concerning the retinal and biomarker changes in DR.

III. COMPARATIVE ANALYSIS

This section presents the comparative analysis of various models from the literature spanning across various applications of AI in DR discussed in the previous section. The analysis is done considering the performance metric Accuracy. Three bar charts are given for ML, DL, XAI. Figure 12 shows the comparison of performance of various ML models used for DR. from the graph it can be seen that decision tree performs better than all the other models found in the literature considered for this survey.

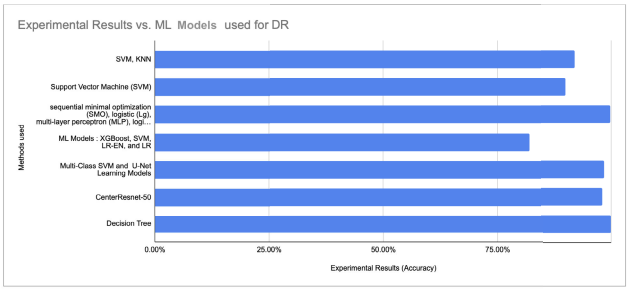


FIGURE 12. Comparative analysis of ML models used for DR.

Figure 13 shows the comparison of performance of various DL models used for DR. from the graph it can be observed that ADL-CNN model and ICOA-DLDRD model outperform all the other DL models from the literature.

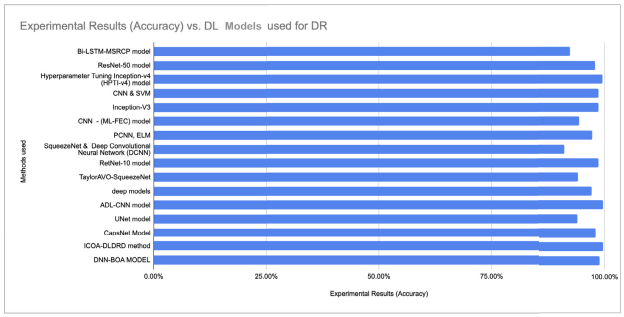


FIGURE 13. Comparative analysis of DL models used for DR.

Figure 14 shows that Hybrid FD-CNN architecture is the best performing one out of all the other models from the literature seen in section IV.

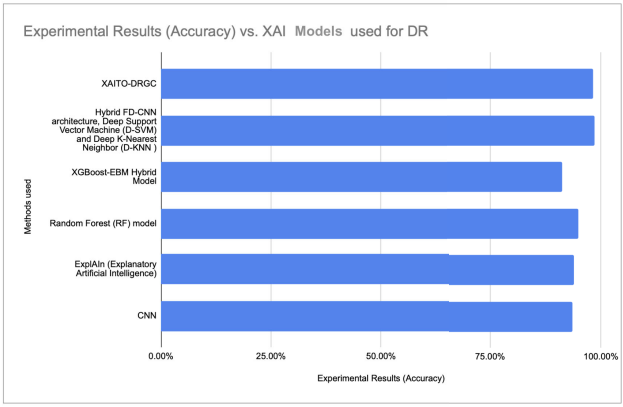


FIGURE 14. Comparative analysis of XAI models used for DR.

IV. DR CASE STUDY

This section presents the significant case studies leveraging the AI techniques in solving the DR problem.

A comprehensive review of recent advancements in DR screening across different countries showcasing

TABLE 10. Summary of DR grading techniques.

Application	Ref. Paper	Method	Dataset	Performance Metrics	Limitation
Grading	[113]	DR-ResNet+	Kaggle Diabetic Retinopathy Detection	Accuracy 98.98%	NA
	[114]	EfficientNet batch normalization (BNs)	EyePACS and DeepDRiD	F1-scores above 80%	The study did not employ complex preprocessing, data augmentation techniques, or additional feature integration in the models.
	[115]	Inceptionv4 model	One thousand seven hundred ninety-six retinal fundus images from 1612 diabetic patients	Mean Sensitivity 88.7%	The methodology exhibits variability, sometimes being overly assertive or overly cautious in highlighting specific image regions.

the application of AI techniques are presented by Bellemo et al. [116]. They utilized datasets such as EyePACS-1, MESSIDOR-2, and E-Ophtha for their experimental analysis. Their experimental findings have shown that in India, the AI software exhibited a specificity of 80% and sensitivity of 96% for discovering any form of DR, and a specificity of 80% and sensitivity of 99% for identifying DR that could pose a threat to eye vision. Similarly, in Thailand, a DL system achieved a sensitivity of 97% in identifying DR, outperforming regional graders who attained a sensitivity of 74%. Looking ahead, the authors suggest that breaking down classification tasks into distinct intermediate objectives could lead to improved performance.

Zhang et al. [117] conducted an experimental study to validate an AI-driven DR screening method and investigated the DR occurrence among adult diabetes patients in China. The research was conducted prospectively across 155 diabetes centers nationwide. A single non-mydriatic fundus photograph focusing on the macula was obtained per eye and assessed using a DL-based classification system with five stages for DR. The initial dataset comprised 143,626 FIs from 37,231 patients collected between 2005 and 2015 from a substantial private RI database. The second dataset included 1,184 color FI from a Chinese public hospital, each graded for DR severity by consensus among three ophthalmologists. The DR grading algorithms exhibited a sensitivity of 83.3% (95% CI- 81.9% to 84.6%) and a specificity of 92.5% (95% CI- 92.1% to 92.9%) in perceiving the DR stages. The performance of the 5-stage DR classification (concordance-83%) was analogous to inter-observer unpredictability among experts (concordance-84.3%) Limitations of the study included the inability to fully reflect the prevalence of DR in the broader population due to its multicenter nature, and the comparatively lower estimated prevalence of DR and referable DR based on the reference standard in a subset of randomly selected participants compared to those identified through AI screening. Future work aims to minimize the proportion of low-quality images to improve the reliability of DR grading.

Another case study was conducted by Xie et al. [118] aimed at evaluating potential cost savings associated with two different deep-learning approaches through a cost-minimization analysis. The models utilized in this research were the TreeAge Pro and decision tree model. The SiDRP dataset, managed by a steering committee composed of representatives from the Singapore Health Service and National Healthcare Group, was used in this study. The experimental results indicated that the semi-automated and fully automated models achieved cost savings of $15(adecreaseof 19.5\%)and 11$ (a decrease of 14.3%) per patient, respectively, when compared to the human assessment model. However, the study acknowledged several limitations. These include the inability to generalize the findings to different contexts, the exclusion of cases involving DME with mild referable DR, and the use of real data and referral guidelines specific to Singapore, which may introduce some referral bias.

Lee et al. [119] conducted a non-interventional device validation study at multiple centers. The study aimed to evaluate the effectiveness of seven automated screening algorithms for DR, including one that is FDA-approved. The algorithms were compared to human graders using real-world retinal imaging data. The dataset used in the study consisted of 311,604 RI collected between 2006 and 2018. These images were obtained from 23,724 veterans who underwent teleretinal screening for DR at either the Veterans Affairs (VA) Puget Sound Health Care System (HCS) or Atlanta VA HCS. Among the methods evaluated, two exhibited higher sensitivities, while one showed sensitivities and specificities similar to others, with a sensitivity of 80.47% ($p = 0.441$) and a specificity of 81.28% ($p = 0.195$). This study identified several limitations, including the predominance of elderly male veterans (94.68%) diagnosed with type 2 DM, reliance of the semi-automated system on algorithms for operation, lack of differentiation between mild and severe levels of DR in the VA system's referable DR threshold, and the use of expert human graders as the reference standard for evaluating algorithm performance compared to tele-retinal

TABLE 11. XAI approaches for DR classification.

Ref. Paper	Method Used	Dataset Used	Limitation
[120]	Classifier: DFNN, Explainer: LIME	MESSIDOR	Assumes static features, missing DR's dynamic changes, generalization issues, and hardware challenges limit the study.
[38]	Classifier: RF model, Explainer: SHAP	Clinical datasets were collected from SRM Hospital 172 individuals with type II diabetes.	Datasets with other health parameters are not taken into account.
[110]	Classifier: FD-CNN, Explainer: LIME	Duke OCT and University of California San Diego (UCSD) datasets	NA
[111]	Classifier: XGBoost Explainer: EBM	Publicly Accessible Dataset: 317 T2D patients with varying retinopathy.	NA

graders. Thus, these case studies outlines the conventional mechanisms applied in DR screening versus AI-based screening techniques, in particular to diabetes patients prone to eye vision threat.

V. EXPLAINABLE ARTIFICIAL INTELLIGENCE FOR DIABETIC RETINOPATHY

This section presents a brief view on XAI for DR. XAI improves the transparency of AI models used for diagnosing DR by elucidating how decisions are reached based on RI.

This is accomplished by employing different frameworks such as LIME, SHAP, and GRADCAM to interpret the models. The XAI methods used in the works referred to are as follows LIME [110], [120], SHAP [38], EBM [111].

XAI provides several benefits in the diagnosis of DR, it identifies biases to ensure equitable diagnosis and improves patient comprehension with clear explanations, vital for chronic conditions such as DR. XAI also aids regulatory adherence and drives research by uncovering new biomarkers, thereby enhancing diagnostic precision and patient care results.

Hence, XAI in healthcare encompasses technologies that enable practitioners to grasp the reasoning behind the system's decisions, essentially elucidating the underlying logic of the AI system.

Table 11 provides information on the XAI research works conducted for DR. XAI is used by various authors for the enhanced meaningful predictions to be made by the DL models. Majority of the works focus on combining DL with XAI for making informed decisions.

An Explainable DL Model for accurate DR severity assessment is developed by Kumar et al. [120]. The model is trained on a publicly accessible dataset MESSIDOR. The AHE, Gamma Correction, and CLAHE techniques preprocess the FI in the dataset. Feedforward Neural Network is used for classification and a Local Interpretable Model-agnostic Explanation (LIME) is used as an explainer. The model has given a sensitivity of 99.69%, accuracy of 98.04%, error rate of 3.60%, f 1-score of 96.99%, and specificity of 96.37%. One limitation is assuming that features learned at different neural network levels remain consistent across all cases, potentially overlooking active modification as time lapses in DR. This constraint can hinder the model's ability to generalize to real-world external datasets. Another challenge lies in potential limitations posed by computer hardware capabilities. Future research could explore the integration of multiple artificial neural networks into a collaborative framework for acquiring data with different features. Furthermore, methods for dynamically identifying pertinent features from feedback data, continually informing the ensemble with updated information, and formulating newer rule-based systems that could potentially improve the complete system working environment.

A feasibility study is done by Lalithadevi et al. [38] for early diagnosis and prevention of DR, a potentially fatal disease for those with diabetes. SHAP XAI technologies were used to apply the RF model. A feasibility study involves the clinical datasets collected from SRM hospital to examine the DR risk in 172 people with type II diabetes. The authors have Added missing values, separated the dataset into validation and discovery sets, and tuned hyperparameters using Bayesian optimization. The analysis's prediction of DR prevalence with an RF model yielded a high accuracy of 94.9%. A better understanding of feature significance was made possible by the SHAP XAI tools. The proposed model suffers due to the limitation that the sample collected is restricted to Type-II diabetic patients, less sample size, along

with other parameters such as urologic, cardiovascular, and neurological ailments aren't considered.

This study examined various socio-analytical, diabetes-related, behavioral, and clinical risk factors in patients with type II diabetes. They have conducted individual questionnaire sessions with all participants, focusing on aspects such as the reliability of results, comfort levels during the screening test, willingness to engage in future screenings, overall impressions, and satisfaction with the DR screening process, and then the authors utilized SHAP (SHapley Additive exPlanations) tools, a robust XAI method, to provide deeper insights into how the input variables influence their corresponding outputs and to rank these input variables. Shapley values can be computed in two main ways: through coalitions and paths. In the coalition approach, features of a dataset are grouped into multiple overlapping subsets known as coalitions. For each coalition, a prediction is made using only the features within that subset, which is then compared to the prediction made when an additional feature is included. In the path approach, calculating Shapley values can be viewed as a dynamic programming problem, where various steps or paths are created. Each step in a path involves incrementally adding another feature from the dataset to produce the prediction. By generating numerous paths, we can explore the different sequences in which features might be included.

OCT images in combination with a Fully Dense-Fusion-NeuralNetwork is used by Kayadibi et al. [110] along with dual preprocessing approaches, and conventional ML methods to enhance the early identification of retinal diseases. The suggested model uses OCT images to diagnose retinal diseases more transparently by combining a Hybrid Fully Dense (FD)-CNN architecture with CNN, ML techniques (D-KNN, D-SVM), and the XAI approach LIME. The FD-CNN performs significantly better on OCT images when dual preprocessing is used, which uses a speckle reduction filter (HSR) with anisotropic diffusion, bilateral, and Gaussian filters. The Duke OCT and University of California San Diego (UCSD) datasets are used to test the models. D-SVM performed well in both datasets with a sensitivity of 98.62%, average accuracy of 98.55%, specificity of 99.39%, F1 score of 98.82%, and precision of 98.60%, indicating the effectiveness of the recommended hybrid approach in automatic classification of retinal illnesses using OCT images. More XAI techniques for transparency, improved preprocessing, enlarged datasets, and an evaluation of the suggested model's clinical applicability might all be explored in future studies.

XAI is used by Yagin et al. [111] to accurately diagnose DR. The model predicts DR subclasses accurately by integrating an Explainable Boosting Machine (EBM) with eXtreme Gradient Boosting (XGBoost). The study utilized open-access data from a total of 317 patients with Type 2 Diabetes (T2D), comprising 143 patients with NDR, 51 patients with PDR, and 123 patients with NPDR. The dataset consists of 317 samples and 145 features. Initially, 39 features had missing values, which were imputed by replacing them with the respective mean values of each feature. Hence, the dataset was fragmented into discovery and validation sets, and the samples were randomly assigned for model development and evaluation, respectively. The latter is separated into discovery and validation sets and aids in feature selection and hyperparameter tuning. To guarantee model performance, the discovery set was considered and a cross-validation was conducted. The data is split into 90% discovery sets and 10% validation sets for preprocessing. XGBoost in conjunction with EBM feature selection yields the best model performance when hyperparameter adjustment is done via Bayesian optimization. Impressive measures, including, 91.24% recall, 89.37% F1-Score,91.25% accuracy, 89.33% precision, and 97.00% AUROC, are obtained from the evaluation of type 2 diabetic patients. The study calls for further examinations and multicenter cohort studies in light of the constraints of external validity. Multi-omic data should be used in future studies to enhance predictions and provide thorough external validation.

VI. CHALLENGES & FUTURE SCOPE

Though many works have been conducted in the existing literature, it becomes evident that numerous challenges persist in the DR application. The Following discusses the significant key challenges found widely in adopting the DR application.

The dataset quality significantly impacts model performance, and limited dataset availability poses additional challenges. Studies often focus narrowly on specific conditions or impose constraints, such as selecting exactly 30 features, which can affect model complexity and performance [38], [85]. Low-dimensional datasets risk performance degradation and overfitting, necessitating careful validation and management of model complexity [107]. Data quality impacts DR diagnosis as AI models rely heavily on training data. A study [114] did not use advanced preprocessing, data augmentation, or additional feature integration, potentially affecting model generalization and handling complexities. Another study highlighted class imbalance in fundus datasets, which can hinder accurate prediction of underrepresented classes [64]. The model's performance can be impacted by prolonged training times due to large datasets, optimization, and hyperparameter tuning [86]. Using ACO during QRHMI parameter optimization led to time consumption issues [65], and trial-and-error hyperparameter setting also contributed to increased training time [70]. Though the model achieves better performance, the applicability of the model in the real-time clinical environment stands pivotal. Due to the dataset biases and diverse image qualities, model needs the clinical validation [86]. The study [70] lacks a clinically acquired database which makes it unreliable for the clinical application.

There is a scope for future research in the realm of AI for DR, where the work can be focussed on data preprocessing,

for a better performance of the model bias, diverse image processing methods can be investigated, and work can be carried out on illumination correction methods. Advanced segmentation techniques must be devised to identify the intricate complex structure effectively. The multi-modal data analysis needs to be addressed so that the model prediction can be enhanced. Furthermore, the data augmentation techniques concerning the data modalities must be incorporated such that the data can be made complete and more insightful. More appropriate model architectures that are tailored for this specific purpose are needed such that the detection can be done with utmost precision. Multimodal data fusion involves integrating multiple modalities to create a comprehensive multimodal representation. There are several key considerations to keep in mind when fusing different modalities [121], [122]: **Intermodality:** This refers to the integration of various modalities, which can enhance model predictions and improve their robustness [123]. **Cross-modality:** This aspect emphasizes the essential interactions between modalities, indicating that meaningful insights cannot be obtained unless all relevant modalities are combined [124]. **Missing data:** In some situations, certain modalities may be unavailable. An effective multimodal data fusion algorithm should be resilient to the absence of some modalities, utilizing the available ones to mitigate the loss of information, particularly in recommender systems. Traditionally, multimodal representation techniques are categorized into two types: early (feature) fusion and late (decision) fusion. In the early fusion approach, all modalities are combined by concatenating their vector representations at an initial stage, followed by training a single model. In contrast, late fusion involves training multiple independent models for each modality and then integrating their outputs. This integration can be done in various ways, such as averaging the outputs or selecting the most common result in classification tasks, or by concatenating the outputs to create a new model for the final output. Neither approach is definitively superior; both have demonstrated promising results across different scenarios [125], [126]. The multimodal fusion is usually done by DL, hashing ideas, sketch representation. The most widely used multimodal fusion techniques rely on DL approaches. Though the promising results were observed by the application of DL in the field of multimodal data fusion, DL models suffer from two main issues Firstly, DL models have a large number of free parameters, particularly those linked to modalities that contribute minimal information, leading to significant resource demands—an undesirable aspect in production settings. Secondly, multimodal data often originate from highly dynamic environments, necessitating a flexible model that can rapidly adapt to changes in the data.

VII. CONCLUSION

DR is gaining its importance as it opens avenues to overcome the limitations of model complexity and generality by leveraging the potential of AI technologies in understanding diagnosing complex conditions. Analyzing the complex real-time patient data such as clinical scores, imaging techniques and wearable sensor reveal complex relationships and patterns that researchers for model generalization and interpretation may overlook. The retinal FI are conventional source of eye images that are used for the diagnosis of occular diseases, but the microscopic details of retinal internal structure cannot be fetched from the FI. Whereas OCT images provide more clear view of the retinal structure compared to FI which can enhance the diagnostic accuracy. Currently, researchers are leveraging the power of AI technologies for DR applications to improve the model performance and trustworthiness. This review article has comprehensively discussed the current research works utilizing the AI technologies for DR applications by presenting the available benchmarking datasets, performance metrics, key challenges faced by the researchers in the field of DR, and future avenues for clinical adoption. It also presents few real-time case studies leveraging the AI technologies in the field of DR. This review can be used to identify the best-performing AI/ML model for various purposes such as classification, segmentation, grading, early detection, early diagnosis and can be further extended with a special focus on multi-modal data for DR and image processing techniques in particular for the image modality which can provide new insights into the practical challenges with respect to the conventional data inputs used most often. This comprehensive review endeavors to navigate the complexities of trustworthy model for clinical validation offering valuable insights into leveraging the full potential of AI in the field of various DR applications.

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